

QRISK_meta_analysis

```
install.packages("readxl") install.packages("meta") install.packages("metafor") install.packages("dplyr") in-
stall.packages("ggplot2")
```

IMPORT DATA FROM EXCEL FILE

```
# Import the data from the xcel fille
# Load package
library(readxl)

# Google Drive file ID (uploaded .xlsx, not a native Sheet)
sheet_id <- "1NjldJRQ_qM6wM4eAfa2cxahBVpHxiB25"
url <- paste0("https://drive.google.com/uc?export=download&id=", sheet_id)

tmp <- tempfile(fileext = ".xlsx")
download.file(url, tmp, mode = "wb")      # mode = "wb" is essential
df <- read_excel(tmp)

# View first few rows
head(df)

## # A tibble: 6 x 41
##   Number Index Author   Title Sample_size Study_Year Disease_Group QRISK_Version
##   <dbl> <chr> <chr>   <chr>      <dbl>      <dbl> <chr>          <chr>
## 1     1  1  Enguita~ 01 E~      20793      2025 "Diabetes (T~ QRISK2
## 2     3  3  Azoulay~ 03 P~       143      2025 "Systemic Lu~ QRISK3
## 3     4  4  Krishna~ 04 C~      5371      2024 "Diabetes (T~ QRISK3
## 4     5  5a  Llorca ~ 05 P~       708      2012 "\r\n Rheuma~ QRISK3
## 5     5  5b  Llorca ~ 05 P~       692      2012 "Ankylosing ~ QRISK3
## 6     5  5c  Llorca ~ 05 P~       680      2012 "Psoriatic A~ QRISK3
## # i 33 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## # `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## # Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## # Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## # QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## # Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## # Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...

# Check structure
str(df)

## tibble [34 x 41] (S3: tbl_df/tbl/data.frame)
## $ Number      : num [1:34] 1 3 4 5 5 5 6 6 6 7 ...
```

```

## $ Index : chr [1:34] "1" "3" "4" "5a" ...
## $ Author : chr [1:34] "Enguita-German et al. (T2DM-2025)" "Azoulay et al. (SLE-2025)"
## $ Title : chr [1:34] "01 External validation of cardiovascular risk scores?in patien
## $ Sample_size : num [1:34] 20793 143 5371 708 692 ...
## $ Study_Year : num [1:34] 2025 2025 2024 2012 2012 ...
## $ Disease_Group : chr [1:34] "Diabetes (T2DM)" "Systemic Lupus Erythematosus (SLE)" "Diabet
## $ QRISK_Version : chr [1:34] "QRISK2" "QRISK3" "QRISK3" "QRISK3" ...
## $ Outcome : chr [1:34] "CV death, non-fatal MI, non-fatal stroke" "The primary outcom
## $ AUC : chr [1:34] "0.663 (0.646 - 0.681)" "0.76 (0.57-0.93)" "0.788 (0.713-
0.863)" "0.60660000000000003" ...
## $ Calibration : chr [1:34] "CITL men and women -1.292 -1.455 (type 1 recalibration) \r
Underestimated.\r\nPsA - Underestimated\r\nAS - Underestimated\r\nHR 1.04 (1.03-1.05)" ...
## $ Follow-Up : chr [1:34] "5 years" "7 years (IQR 3-9 years)" "1 - 12 years\r\nLongitudin
## $ Country : chr [1:34] "Spain" "France" "India" "Spain" ...
## $ Participants : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ Predictors2 : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ Outcome3 : chr [1:34] "HIGH" "HIGH" "HIGH" "LOW" ...
## $ Analysis : chr [1:34] "LOW" "HIGH" "HIGH" "UNCLEAR" ...
## $ Overall : chr [1:34] "HIGH" "HIGH" "HIGH" "UNCLEAR" ...
## $ Metanalysis : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
## $ Confidence_interval : chr [1:34] "Pooled" "Reported" "Approximated" "Not_reported" ...
## $ Dscrimination : chr [1:34] "Harrell's C-index" "AUC" "AUC" "Heller's K" ...
## $ QRISK_2 : num [1:34] 1 0 0 0 0 0 0 0 0 0 ...
## $ QRISK_3 : num [1:34] 0 1 1 1 1 1 1 1 1 1 ...
## $ Outcome_type : chr [1:34] "Appropriate" "Appropriate" "Somewhat_Appropriate" "Appropriat
## $ Outcome_Description : chr [1:34] "CVD" "CVD" "CVD( not including peripheral arte-\r\nrial disea
## $ Applicability : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
## $ Include : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
## $ PA : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ PR : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ OT : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ OA : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
## $ Study_2 : chr [1:34] "01 Enguita-German" "03 Azoulay 2025 (Lupus)" "04 Aswath Krishn
## $ Single_vs_multi-centre: chr [1:34] "Multi-centre" "Single-centre" "Single-centre" "Multi-centre"
## $ Events : num [1:34] 991 12 50 67 50 57 607 70 485 34 ...
## $ Mean_age : chr [1:34] "63.5" "51" "Not reported for whole cohort (low-risk 38.2, high
## $ SD_age : chr [1:34] "10.8" "Not provided" "Not reported for whole cohort (low-risk
## $ Male : chr [1:34] "12225" "13" "486" "Not reported by disease subgroup (whole coh
## $ Female : num [1:34] 8568 130 4885 NA NA ...
## $ Sensitivity_clean : chr [1:34] "Not provided" "83%" "0.78" "Not provided" ...
## $ Specificity_clean : chr [1:34] "Not provided" "71%" "0.69899999999999995" "Not provided" ...
## $ Calibration_shot : chr [1:34] "OVER-estimation of risk in original model; good calibration on

```

```

# Check column names
names(df)

```

```

## [1] "Number" "Index" "Author"
## [4] "Title" "Sample_size" "Study_Year"
## [7] "Disease_Group" "QRISK_Version" "Outcome"
## [10] "AUC" "Calibration" "Follow-Up"
## [13] "Country" "Participants" "Predictors2"
## [16] "Outcome3" "Analysis" "Overall"
## [19] "Metanalysis" "Confidence_interval" "Dscrimination"

```

```
## [22] "QRISK_2"          "QRISK_3"          "Outcome_type"
## [25] "Outcome_Description" "Applicability"    "Include"
## [28] "PA"              "PR"              "OT"
## [31] "OA"              "Study_2"         "Single_vs_multi-centre"
## [34] "Events"          "Mean_age"        "SD_age"
## [37] "Male"            "Female"          "Sensitivity_clean"
## [40] "Specificity_clean" "Calibration_shot"
```

CLEAN THE IMPORTED DATA BY REMOVING ANY EMPTY COLUMNS THE RECHECK DATA

```
# Remove columns that are entirely NA
df <- df[, colSums(!is.na(df)) > 0]

# Check dimensions
dim(df)
```

```
## [1] 34 41
```

```
# Check column names
names(df)
```

```
## [1] "Number"          "Index"            "Author"
## [4] "Title"           "Sample_size"      "Study_Year"
## [7] "Disease_Group"   "QRISK_Version"    "Outcome"
## [10] "AUC"             "Calibration"      "Follow-Up"
## [13] "Country"         "Participants"     "Predictors2"
## [16] "Outcome3"        "Analysis"         "Overall"
## [19] "Metanalysis"     "Confidence_interval" "Discrimination"
## [22] "QRISK_2"         "QRISK_3"          "Outcome_type"
## [25] "Outcome_Description" "Applicability"    "Include"
## [28] "PA"              "PR"              "OT"
## [31] "OA"              "Study_2"         "Single_vs_multi-centre"
## [34] "Events"          "Mean_age"        "SD_age"
## [37] "Male"            "Female"          "Sensitivity_clean"
## [40] "Specificity_clean" "Calibration_shot"
```

REMOVE COMPLETELY EMPTY ROWS

```
# Remove rows where all values are NA
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
df <- df %>%
  filter(!if_all(everything(), is.na))
```

CHECK THE FORMAT OF INPUTED CONFIDENCE INTERVAL IN ORDER TO DEVICE PLAN TO PARSE IT

```
# check CI format
df$AUC[!is.na(df$AUC)][1:20]
```

```
## [1] "0.663 (0.646 - 0.681)" "0.76 (0.57-0.93)"
## [3] "0.788 (0.713-0.863)" "0.60660000000000003"
## [5] "0.59189999999999998" "0.57999999999999996"
## [7] "0.74 (0.72-0.76)" "\r\n0.70 (0.64-0.76)"
## [9] "0.716 (0.69-0.74)" "0.703(0.602 - 0.804)"
## [11] "\r\n\r\n\r\n0.752 (0.734-0.777)" "\r\n\r\n0.794 (0.764-0.812)"
## [13] "0.764 (0.741-0.791)" "\r\n\r\n0.815 (0.789-0.835)"
## [15] "0.754 (0.64-0.87)" "0.878 (0.839 - 0.917)"
## [17] "0.765 (0.711-0.820)" "0.664 (0.660 - 0.667)"
## [19] "0.664 (0.660 - 0.668)" "0.697 (0.659-0.736)"
```

SPLIT CONFIDENCE INTERVAL

CLEAN AND EXTRACT NUMERICAL VALUES FROM AUC COLUMN

```
# extract CI into lower and upper column
library(stringr) # Provides fuction for manipulating text
library(dplyr) # provides function for manipulating data

df <- df %>% # use the pipe operator to pass the result of manipulation and passes it to
  ↪ the right
  mutate( # create new columns while retaining exisiting ones
    AUC_clean = str_squish(AUC), #remove unnecessary spaces from AUC and store it in new
    ↪ variable

    Estimate = str_extract(AUC_clean, "^\\d*\\.?\\d+") |> as.numeric(), # creates a
    ↪ variabl, Estimate and store the first number seen in the AUC_clean is it as a
    ↪ number

    LowerCI = str_match( # Using regular expression extract the lower confidence interval
    ↪ and store it in LowerCI column
      AUC_clean,
      "\\((\\s*(\\d*\\.?\\d+)\\s*[-]\\s*(\\d*\\.?\\d+)\\s*\\)"
    )[,2] |> as.numeric(),

    UpperCI = str_match( # Using regular expression extract the Upper confidence interval
    ↪ and store it in UpperCI column
      AUC_clean,
      "\\((\\s*(\\d*\\.?\\d+)\\s*[-]\\s*(\\d*\\.?\\d+)\\s*\\)"
    )[,3] |> as.numeric()
  )
```

```
# view all the new extracted data
df %>%
  select(AUC, Estimate, LowerCI, UpperCI) %>%
  head(30)
```

```
## # A tibble: 30 x 4
##   AUC                Estimate LowerCI UpperCI
##   <chr>              <dbl>   <dbl>   <dbl>
## 1 "0.663 (0.646 - 0.681)" 0.663 0.646 0.681
## 2 "0.76 (0.57-0.93)"    0.76 0.57 0.93
## 3 "0.788 (0.713-0.863)" 0.788 0.713 0.863
## 4 "0.60660000000000003" 0.607 NA     NA
## 5 "0.59189999999999998" 0.592 NA     NA
## 6 "0.57999999999999996" 0.58  NA     NA
## 7 "0.74 (0.72-0.76)"    0.74 0.72 0.76
## 8 "\r\n0.70 (0.64-0.76)" 0.7   0.64 0.76
## 9 "0.716 (0.69-0.74)"   0.716 0.69 0.74
## 10 "0.703(0.602 - 0.804)" 0.703 0.602 0.804
## # i 20 more rows
```

HANLEY & MCNEIL (1982) APPROXIMATION

```
# HANLEY & MCNEIL (1982) APPROXIMATION
library(dplyr)

df_hm <- df # create a working copy

df_hm <- df_hm %>%
  mutate( # create a new column
    n1 = as.numeric(Events), # store the information in the Event column in n1 as number
    n0 = Sample_size - n1, # compute the number of none events by subtracting event from
    # sample size
    Q1 = Estimate / (2 - Estimate), # compute for Q1
    Q2 = 2 * Estimate^2 / (1 + Estimate), # compute for Q2
    HM_se = sqrt((Estimate * (1 - Estimate) + # h and M original variance estimator
      (n1 - 1) * (Q1 - Estimate^2) +
      (n0 - 1) * (Q2 - Estimate^2)) / (n1 * n0)),
    HM_lower = pmax(0, Estimate - 1.96 * HM_se), # calculate lower confidence interval
    HM_upper = pmin(1, Estimate + 1.96 * HM_se) # calculate higher confidence interval
  )

# Output the values that were affected by the approximation
hm_table <- df_hm %>% # what would be
  filter(Confidence_interval == "Approximated") %>% # isolate row that currently do not
  # have reported confidence interval
  transmute(
    Study = Index, # indentify the row by the index values
    Events = n1,
    AUC = round(Estimate, 3),
    `95% CI` = sprintf("%.3f - %.3f", HM_lower, HM_upper)
  )
```

```
print(as.data.frame(hm_table), row.names = FALSE)
```

```
## Study Events AUC 95% CI
## 4 50 0.788 0.713 - 0.863
## 7 34 0.703 0.602 - 0.804
## 14 26 0.754 0.639 - 0.869
## 16 138 0.878 0.839 - 0.917
## 26a 232 0.720 0.681 - 0.759
## 26b 232 0.670 0.630 - 0.710
## 32 9 0.510 0.317 - 0.703
```

CALCULATE STANDARD ERROR

```
# STANDARD ERROR
df$SE <- (df$UpperCI - df$LowerCI) / 3.92
```

```
# VERIFY NO VLAUES ARE MISSING
sum(is.na(df$SE))
```

```
## [1] 3
```

```
summary(df$SE)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max. NA's
## 0.001786 0.011990 0.020408 0.032039 0.040816 0.096939 3
```

SPLIT DATA (df_all, df_primary)

df_all : dataframe containing all included studies

df_primary : dataframe containing only studies included in the primary pool

```
# Load the package
library(writexl)
```

```
# create df with all data including QRISK
df_all <- df

nrow(df_all)
```

```
## [1] 34
```

```
head(df_all)
```

```
## # A tibble: 6 x 46
##   Number Index Author   Title Sample_size Study_Year Disease_Group QRISK_Version
##   <dbl> <chr> <chr>   <chr>      <dbl>      <dbl> <chr>      <chr>
## 1     1 1     Enguita~ 01 E~      20793      2025 "Diabetes (T~ QRISK2
## 2     3 3     Azoulay~ 03 P~        143      2025 "Systemic Lu~ QRISK3
## 3     4 4     Krishna~ 04 C~      5371      2024 "Diabetes (T~ QRISK3
## 4     5 5a    Llorca ~ 05 P~        708      2012 "\r\n Rheuma~ QRISK3
## 5     5 5b    Llorca ~ 05 P~        692      2012 "Ankylosing ~ QRISK3
## 6     5 5c    Llorca ~ 05 P~        680      2012 "Psoriatic A~ QRISK3
## # i 38 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## #   `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## #   Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## #   Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## #   QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## #   Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## #   Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...
```

```
# Export the dataframe
write_xlsx(df_all, "df_all.xlsx")
```

Extracting the primary pool studies

```
# create dataframe for primary pooled AUC
# keep only studies marked for inclusion
df_primary <- df[df$Include == 1, ]

# keep only studies marked for meta - analysis
df_primary <- df_primary[df_primary$Metanalysis == 1, ]

# remove hellers k studies
df_primary <- df_primary[df_primary$Dscrimination != "Heller's K", ]

df_primary <- df_primary[df_primary$Outcome_type == "Appropriate", ]

# View first rows
nrow(df_primary)
```

```
## [1] 19
```

```
str(df_primary$Index)
```

```
## chr [1:19] "1" "3" "6a" "6b" "6c" "7" "8a" "8b" "8c" "8d" "16" "19b" "23" ...
```

```
write_xlsx(df_primary, "df_primary.xlsx")
```

TRANSFORM EACH C-STATISTIC TO LOGIT SCALE FOR DEBRAY'S ADHERENCE

```
# transform the AUC values from their original probability scale (0-1) to the logit scale
↪ and calculates the standard error (SE) on the logit scale.
library(dplyr)
```

```
df_primary_logit <- df_primary %>%
  mutate(
    logit_c = log(Estimate / (1 - Estimate)), # create logit transformed AUC
    logit_lb = log(LowerCI / (1 - LowerCI)),
    logit_ub = log(UpperCI / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96) # derive the standard error of the
    ↪ logit c-statistic
  )
```

```
# Export the dataframe
write_xlsx(df_primary_logit, "df_primary-logit.xlsx")
```

```
# TO CHECK
# output unique studies for primary metanalysis
library(dplyr)
library(stringr)

# Eligible cohorts
eligible <- df_primary_logit %>%
  filter(Metanalysis == 1) %>%
  mutate(
    StudyID = str_extract(as.character(Index), "\\d+")
  )

nrow(eligible)
```

```
## [1] 19
```

```
cat(
  "Number of unique studies:",
  n_distinct(eligible$StudyID),
  "\n"
)
```

```
## Number of unique studies: 14
```

```
eligible %>%
  select(Index, Title) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 2
##   Index Title
##   <chr> <chr>
## 1 1      01 External validation of cardiovascular risk scores?in patients with ~
## 2 3      03 Performance of SCORE2, QRISK3 and PREVENT equations in systemic lup~
## 3 6a     06 External validation of the accuracy of cardiovascular risk predicti~
## 4 6b     06 External validation of the accuracy of cardiovascular risk predicti~
## 5 6c     06 External validation of the accuracy of cardiovascular risk predicti~
## 6 7      07 Comparison of Cardiovascular Risk Scores Performances in a Contempo~
```



```
## 7 8a    08 The predictive accuracy of cardiovascular disease risk?prediction t~
## 8 8b    08 The predictive accuracy of cardiovascular disease risk?prediction t~
## 9 8c    08 The predictive accuracy of cardiovascular disease risk?prediction t~
## 10 8d   08 The predictive accuracy of cardiovascular disease risk?prediction t~
## 11 16   16 Assessment of QRISK3 as a predictor of cardiovascular disease event~
## 12 19b  19 Cardiovascular risk prediction in type 2 diabetes/ a comparison of ~
## 13 23   23 Electronic Medical Record Risk Modeling of Cardiovascular Outcomes ~
## 14 26b  26 Assessment of cardiovascular risk tools as predictors of cardiovasc~
## 15 29b  29 Cardiovascular Risk Prediction in Ankylosing Spondylitis/ From Trad~
## 16 35   35 Performances of five risk algorithms in predicting cardiovascular e~
## 17 36   36 Performance of cardiovascular disease risk scores in people diagnos~
## 18 39   39 Comparison of cardiovascular risk algorithms in patients with vs wi~
## 19 40   40 Performance of four current risk algorithms?in predicting cardiovas~
```

Count the number of arms contributed by each study

```
df_primary_logit %>%
  mutate(StudyID = str_extract(as.character(Index), "[0-9]+")) %>% # CONVERT index to
  ↪ character and then pick out characters from the begining until you find a character
  ↪ that is not a number and store that in studyID
  count(StudyID, name = "n_arms") %>% # get the frequency of each number and store in the
  ↪ column n_arms
  arrange(desc(n_arms)) %>% # arrange the result in decending order
  print(n = Inf) # print all rows
```

```
## # A tibble: 14 x 2
##   StudyID n_arms
##   <chr>    <int>
## 1 8        4
## 2 6        3
## 3 1        1
## 4 16       1
## 5 19       1
## 6 23       1
## 7 26       1
## 8 29       1
## 9 3        1
## 10 35      1
## 11 36      1
## 12 39      1
## 13 40      1
## 14 7        1
```

META-ANALYSIS STARTS

```
# LOAD META PACKAGE
library(meta)
```

```
## Loading required package: metabook
```

```
## Loading 'meta' package (version 8.5-0).
## Type 'help(meta)' for a brief overview.
```

```
# INVERSE VARIANCE RANDOM EFFECT Using non logit estimates
primary_auc <- metagen( # use metagen package to combine AUC estimates and store in
  → created object
  TE = Estimate, # effect estimate gotten from Estimate column
  seTE = SE, # get the treatment effect standard error from the SE column
  studlab = Author, # use the Author column to identify studies
  data = df_primary, # dataset that contains all the data
  sm = "AUC", # specify the type of summary measure bein meta-analyzed as AUC
  method.tau = "REML", # how the between study variance is estimated
  method.random.ci = "HK" # method for deriving the confidence interval
)
summary(primary_auc)
```

	AUC	95% CI	%W(common)	%W(random)
## Enguita-German et al. (T2DM-2025)	0.6630	[0.6455; 0.6805]	2.6	6.1
## Azoulay et al. (SLE-2025)	0.7600	[0.5800; 0.9400]	0.0	1.9
## Hughes et al. (Ps-2025)	0.7400	[0.7200; 0.7600]	2.0	6.0
## Hughes et al. (PsA-2025)	0.7000	[0.6400; 0.7600]	0.2	5.0
## Hughes et al. (RA-2025)	0.7160	[0.6910; 0.7410]	1.3	5.9
## Rosário et al. (SLE-2025)	0.7030	[0.6020; 0.8040]	0.1	3.6
## Hughes et al. (RA-2023)	0.7520	[0.7305; 0.7735]	1.8	6.0
## Hughes et al. (AS-2023)	0.7940	[0.7700; 0.8180]	1.4	6.0
## Hughes et al. (PsA-2023)	0.7640	[0.7390; 0.7890]	1.3	5.9
## Hughes et al. (Ps-2023)	0.8150	[0.7920; 0.8380]	1.5	6.0
## Mu et al. (T2DM-2022)	0.8780	[0.8390; 0.9170]	0.5	5.6
## Dziopa K et al. (T2DM-Q3-2022)	0.6640	[0.6600; 0.6680]	50.6	6.2
## Hong et al. (T2DM-2021)	0.7200	[0.7000; 0.7400]	2.0	6.0
## Sivakumaran et al. (SLE-Q3-2021)	0.6700	[0.6300; 0.7100]	0.5	5.6
## Navarini et al. (AS-Q3-2020)	0.6600	[0.4800; 0.8400]	0.0	1.9
## Navarini et al. (PsA-2018)	0.8660	[0.7943; 0.9377]	0.2	4.6
## Read et al. (T2DM-2018)	0.6740	[0.6690; 0.6790]	32.4	6.2
## Alemao et al. (RA-2017)	0.7440	[0.7140; 0.7740]	0.9	5.8
## Arts EEA et al. (RA-2013)	0.7900	[0.7500; 0.8300]	0.5	5.6

```
## Number of studies: k = 19
```

	AUC	95% CI	z t	p-value
## Common effect model	0.6805	[0.6777; 0.6834]	468.56	0
## Random effects model (HK)	0.7419	[0.7103; 0.7735]	49.37	< 0.0001

```
## Quantifying heterogeneity (with 95% CIs):
```

```
## tau^2 = 0.0038 [0.0019; 0.0086]; tau = 0.0616 [0.0431; 0.0929]
```

```
## I^2 = 97.0% [96.2%; 97.7%]; H = 5.81 [5.15; 6.54]
```

```
##
```

```
## Test of heterogeneity:
```

```
## Q d.f. p-value
```

```
## 606.94 18 < 0.0001
```

```
##
```

```
## Details of meta-analysis methods:
```

```
## - Inverse variance method
```

```
## - Restricted maximum-likelihood estimator for tau^2
```

```
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
```

```
# POOLED ESTIMATE ON LOGIT SCALE
```

```
library(meta)

primary_logit <- metagen(
  TE = logit_c, # logit effect estimate gotten from logit_c column
  seTE = se_logit, # get the logit treatment effect standard error from the
  ↪ Se_logit column
  studlab = Author, # use the Author column to identify studies
  data = df_primary_logit, # dataset that contains all the logit transformed
  ↪ data
  method.tau = "REML", # how the between study variance is estimated
  method.random.ci = "HK", # method for deriving the confidence interval
  prediction = TRUE # report 95% prediction interval
)
summary(primary_logit)
```

```
##                                     95% CI %W(common)
## Enguita-German et al. (T2DM-2025) 0.6767 [ 0.5983; 0.7551]      2.8
## Azoulay et al. (SLE-2025)         1.1527 [ 0.0003; 2.3051]      0.0
## Hughes et al. (Ps-2025)           1.0460 [ 0.9419; 1.1501]      1.6
## Hughes et al. (PsA-2025)          0.8473 [ 0.5586; 1.1360]      0.2
## Hughes et al. (RA-2025)           0.9247 [ 0.8018; 1.0476]      1.1
## Rosário et al. (SLE-2025)         0.8616 [ 0.3628; 1.3605]      0.1
## Hughes et al. (RA-2023)           1.1093 [ 0.9927; 1.2259]      1.3
## Hughes et al. (AS-2023)           1.3492 [ 1.2050; 1.4934]      0.8
## Hughes et al. (PsA-2023)          1.1747 [ 1.0348; 1.3146]      0.9
## Hughes et al. (Ps-2023)           1.4828 [ 1.3315; 1.6341]      0.8
## Mu et al. (T2DM-2022)             1.9736 [ 1.5979; 2.3493]      0.1
## Dziopa K et al. (T2DM-Q3-2022)    0.6812 [ 0.6632; 0.6991]     53.7
## Hong et al. (T2DM-2021)           0.9445 [ 0.8451; 1.0438]      1.7
## Sivakumaran et al. (SLE-Q3-2021) 0.7082 [ 0.5266; 0.8898]      0.5
## Navarini et al. (AS-Q3-2020)      0.6633 [-0.2058; 1.5324]      0.0
## Navarini et al. (PsA-2018)        1.8660 [ 1.1856; 2.5465]      0.0
## Read et al. (T2DM-2018)           0.7263 [ 0.7036; 0.7491]     33.3
## Alemao et al. (RA-2017)           1.0669 [ 0.9088; 1.2249]      0.7
## Arts EEA et al. (RA-2013)         1.3249 [ 1.0814; 1.5684]      0.3
##                                     %W(random)
## Enguita-German et al. (T2DM-2025) 6.3
## Azoulay et al. (SLE-2025)         1.4
## Hughes et al. (Ps-2025)           6.2
## Hughes et al. (PsA-2025)          5.3
## Hughes et al. (RA-2025)           6.2
## Rosário et al. (SLE-2025)         3.9
## Hughes et al. (RA-2023)           6.2
## Hughes et al. (AS-2023)           6.1
## Hughes et al. (PsA-2023)          6.1
## Hughes et al. (Ps-2023)           6.0
## Mu et al. (T2DM-2022)             4.7
## Dziopa K et al. (T2DM-Q3-2022)    6.4
## Hong et al. (T2DM-2021)           6.2
```

```

## Sivakumaran et al. (SLE-Q3-2021)          5.9
## Navarini et al. (AS-Q3-2020)             2.2
## Navarini et al. (PsA-2018)               2.9
## Read et al. (T2DM-2018)                  6.4
## Alemao et al. (RA-2017)                  6.0
## Arts EEA et al. (RA-2013)                5.6
##
## Number of studies: k = 19
##
##                               95% CI    z|t    p-value
## Common effect model          0.7379 [0.7248; 0.7510] 110.10      0
## Random effects model (HK)    1.0615 [0.8892; 1.2338] 12.94 < 0.0001
## Prediction interval          [0.3746; 1.7484]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 432.04   18 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)

# invert pooled value and intervals with inverse-logit
plogis(primary_logit$TE.random)      #perform inverse logit transformation on the pooled
  ⇨ estimate

## [1] 0.7429772

plogis(c(primary_logit$lower.random, primary_logit$upper.random)) # perform inverse logit
  ⇨ transformation on the confidence interval

## [1] 0.7087175 0.7744896

plogis(c(primary_logit$lower.predict, primary_logit$upper.predict)) # perform inverse
  ⇨ logit transformation in the prediction interval

## [1] 0.5925623 0.8517550

# USING METAMISC LIBRARY AS STATED IN DEBRAY
library(metamisc) # debrey specific package for meta analysis

##
## Attaching package: 'metamisc'

```

```
## The following object is masked from 'package:meta':
##
## forest
```

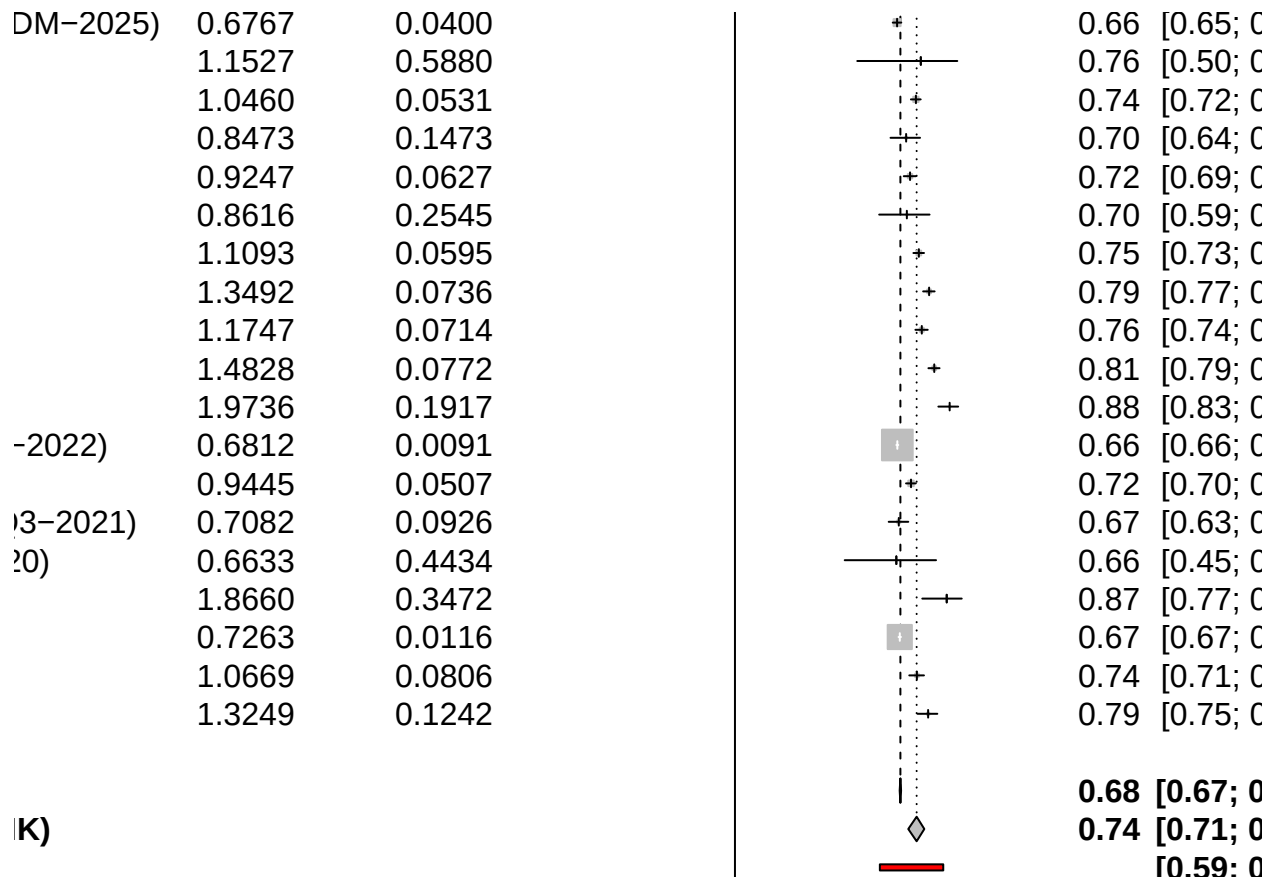
```
val <- valmeta( # store result of the velmeta function in the val object
  cstat      = df_primary_logit$Estimate, # RAW c-statistic provided as logit
  ↪ trransformation is done internally
  cstat.cilb = df_primary_logit$LowerCI, # RAW lower CI
  cstat.ciub = df_primary_logit$UpperCI, # RAW upper CI
  slab       = df_primary_logit$Index, # use index as the study label
  method     = "REML" # how the between study variance is estimated
)
val
```

```
## Summary c-statistic with 95% confidence and (approximate) 95% prediction interval:
##
## Estimate      CIl      CIu      PI1      PIu
## 0.7429770 0.7087174 0.7744894 0.5916327 0.8522402
##
## Number of studies included: 19
```

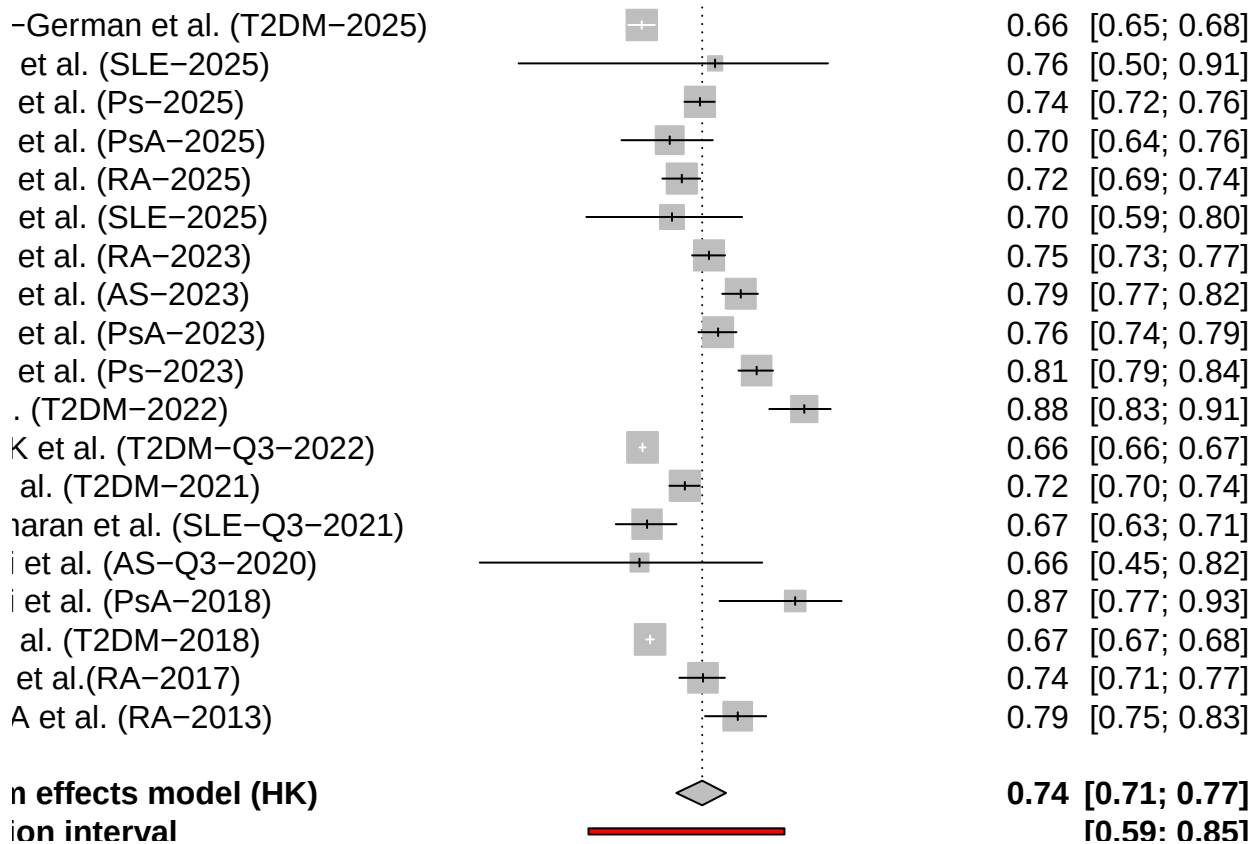
```
# FORREST PLOT VABOSE
library(meta)

primary_logit <- metagen(
  TE      = logit_c, # logit effect estimate gotten from logit_c column
  seTE    = se_logit, # get the logit treatment effect standard error from the
  ↪ Se_logit column
  studlab = Author, # use the Author column to identify studies
  data    = df_primary_logit, # dataset that contains all the logit transformed
  ↪ data
  sm      = "PLOGIT", # tells meta the scale is logit-transformed proportion
  backtransf = TRUE, # covert logit values to translatable values
  method.tau = "REML", # how the between study variance is estimated
  method.random.ci = "HK", # method for deriving the confidence interval
  prediction = TRUE # report 95% prediction interval
)

meta::forest(primary_logit, xlab = "c-statistic") # use forest function in the meta
↪ package
```



```
# PLOT FORREST PLOT
meta::forest(primary_logit,
  xlab = "c-statistic",
  xlim = c(0.4, 1),
  at = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
  leftcols = c("studlab"),
  leftlabs = c("Study"),
  rightcols = c("effect", "ci", "w.random"),
  rightlabs = c("c-statistic", "95% CI", "Weight"),
  common = FALSE,
  print.I2 = TRUE,
  print.tau2 = TRUE)
```



INFLUENCE ANALYSIS

```
#Influence analysis to be done on logit transformed estimate
# result will be back transformed for translation
# starting with leave-one-out sensitivity analysis
# If dropping a study shifts the pooled c-statistic by >0.05 the result is not stable
library(meta)
loo <- metainf(primary_logit, pooled = "random") # leave one out
loo
```

```
## Leave-one-out meta-analysis
##
##
## proportion 95% CI p-value
## Omitting Enguita-German et al. (T2DM-2025) 0.7478 [0.7132; 0.7795] < 0.0001
## Omitting Azoulay et al. (SLE-2025) 0.7428 [0.7070; 0.7756] < 0.0001
## Omitting Hughes et al. (Ps-2025) 0.7435 [0.7067; 0.7771] < 0.0001
## Omitting Hughes et al. (PsA-2025) 0.7454 [0.7095; 0.7783] < 0.0001
## Omitting Hughes et al. (RA-2025) 0.7449 [0.7085; 0.7782] < 0.0001
## Omitting Rosário et al. (SLE-2025) 0.7447 [0.7088; 0.7775] < 0.0001
## Omitting Hughes et al. (RA-2023) 0.7427 [0.7058; 0.7763] < 0.0001
## Omitting Hughes et al. (AS-2023) 0.7394 [0.7033; 0.7725] < 0.0001
## Omitting Hughes et al. (PsA-2023) 0.7418 [0.7051; 0.7754] < 0.0001
## Omitting Hughes et al. (Ps-2023) 0.7374 [0.7024; 0.7697] < 0.0001
```

```

## Omitting Mu et al. (T2DM-2022) 0.7334 [0.7041; 0.7609] < 0.0001
## Omitting Dziopa K et al. (T2DM-Q3-2022) 0.7478 [0.7132; 0.7795] < 0.0001
## Omitting Hong et al. (T2DM-2021) 0.7447 [0.7082; 0.7781] < 0.0001
## Omitting Sivakumaran et al. (SLE-Q3-2021) 0.7472 [0.7122; 0.7792] < 0.0001
## Omitting Navarini et al. (AS-Q3-2020) 0.7447 [0.7095; 0.7770] < 0.0001
## Omitting Navarini et al. (PsA-2018) 0.7382 [0.7051; 0.7689] < 0.0001
## Omitting Read et al. (T2DM-2018) 0.7473 [0.7122; 0.7794] < 0.0001
## Omitting Alemao et al. (RA-2017) 0.7432 [0.7064; 0.7768] < 0.0001
## Omitting Arts EEA et al. (RA-2013) 0.7401 [0.7039; 0.7733] < 0.0001
##
## Random effects model (HK) 0.7430 [0.7087; 0.7745] < 0.0001
## tau^2 tau I^2 95% PI
## Omitting Enguita-German et al. (T2DM-2025) 0.0964 0.3105 96.0% [0.6010; 0.8537]
## Omitting Azoulay et al. (SLE-2025) 0.1025 0.3202 96.1% [0.5899; 0.8529]
## Omitting Hughes et al. (Ps-2025) 0.1096 0.3310 95.7% [0.5848; 0.8564]
## Omitting Hughes et al. (PsA-2025) 0.1057 0.3251 96.1% [0.5904; 0.8561]
## Omitting Hughes et al. (RA-2025) 0.1081 0.3289 96.0% [0.5878; 0.8568]
## Omitting Rosário et al. (SLE-2025) 0.1052 0.3244 96.1% [0.5900; 0.8553]
## Omitting Hughes et al. (RA-2023) 0.1092 0.3304 95.7% [0.5841; 0.8557]
## Omitting Hughes et al. (AS-2023) 0.1010 0.3178 95.3% [0.5866; 0.8502]
## Omitting Hughes et al. (PsA-2023) 0.1079 0.3285 95.7% [0.5840; 0.8547]
## Omitting Hughes et al. (Ps-2023) 0.0917 0.3028 95.0% [0.5919; 0.8446]
## Omitting Mu et al. (T2DM-2022) 0.0665 0.2579 95.6% [0.6104; 0.8285]
## Omitting Dziopa K et al. (T2DM-Q3-2022) 0.0964 0.3105 95.1% [0.6010; 0.8537]
## Omitting Hong et al. (T2DM-2021) 0.1086 0.3296 95.9% [0.5871; 0.8568]
## Omitting Sivakumaran et al. (SLE-Q3-2021) 0.0996 0.3156 96.1% [0.5975; 0.8547]
## Omitting Navarini et al. (AS-Q3-2020) 0.1024 0.3201 96.1% [0.5924; 0.8542]
## Omitting Navarini et al. (PsA-2018) 0.0921 0.3035 96.0% [0.5928; 0.8453]
## Omitting Read et al. (T2DM-2018) 0.0996 0.3156 96.1% [0.5977; 0.8548]
## Omitting Alemao et al. (RA-2017) 0.1093 0.3305 95.9% [0.5847; 0.8561]
## Omitting Arts EEA et al. (RA-2013) 0.1028 0.3206 95.9% [0.5860; 0.8513]
##
## Random effects model (HK) 0.1005 0.3170 95.8% [0.5926; 0.8518]
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = {17, 18})
## - Prediction interval based on t-distribution (df = {17, ..., 18})
## - Logit transformation

meta::forest(loo, backtransf = TRUE, xlab = "c-statistic (leave-one-out)")

```


Study	Forest Plot	OR (95% CI)	P-value
LE-2025)		0.74 [0.71; 0.78]	< 0.000
SA-2023)		0.74 [0.71; 0.78]	< 0.000
AS-2023)		0.74 [0.70; 0.77]	< 0.000
SA-2023)		0.74 [0.71; 0.78]	< 0.000
s-2023)		0.74 [0.70; 0.77]	< 0.000
-2022)		0.73 [0.70; 0.76]	< 0.000
T2DM-Q3-2022)		0.75 [0.71; 0.78]	< 0.000
DM-2021)		0.74 [0.71; 0.78]	< 0.000
al. (SLE-Q3-2021)		0.75 [0.71; 0.78]	< 0.000
AS-Q3-2020)		0.74 [0.71; 0.78]	< 0.000
SA-2018)		0.74 [0.71; 0.77]	< 0.000

```

data = df_primary_logit, # location of variables
method = "REML", # methos for calculating study estimate
slab = df_primary_logit$Index)

inf <- influence(rma_meta) # calcualte the influence statistic for every study

print(inf$inf)

```

```

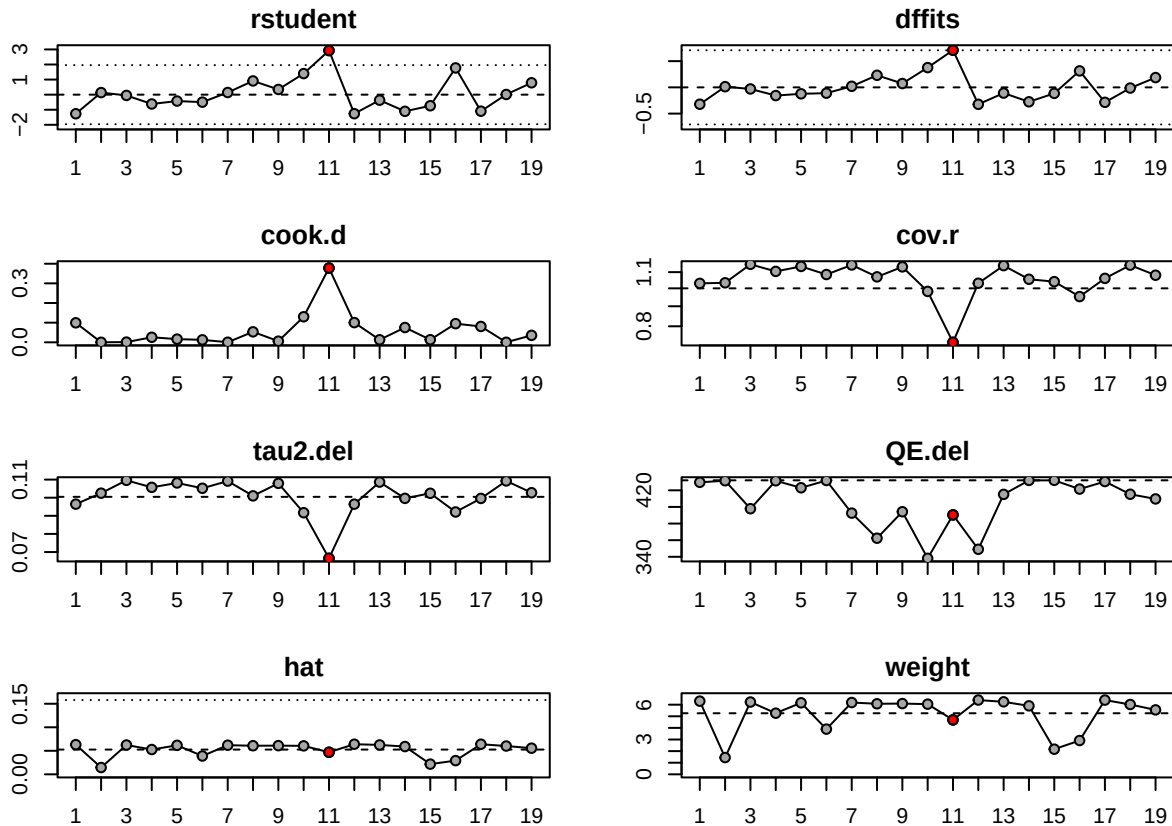
##
##      rstudent  dffits cook.d  cov.r tau2.del   QE.del   hat weight inf
## 1      -1.2676 -0.3214 0.0992 1.0295   0.0964 429.6310 0.0631 6.3053
## 3       0.1367  0.0126 0.0002 1.0331   0.1025 431.5393 0.0144 1.4425
## 6a     -0.0522 -0.0302 0.0010 1.1515   0.1096 397.8521 0.0623 6.2308
## 6b     -0.6190 -0.1566 0.0256 1.1046   0.1057 431.4840 0.0527 5.2685
## 6c     -0.4254 -0.1229 0.0162 1.1374   0.1081 423.0615 0.0616 6.1645
## 7      -0.4965 -0.1105 0.0126 1.0846   0.1052 431.8005 0.0389 3.8948
## 8a      0.1428  0.0203 0.0004 1.1472   0.1092 392.5748 0.0619 6.1878
## 8b      0.9097  0.2303 0.0533 1.0698   0.1010 362.3787 0.0608 6.0785
## 8c      0.3440  0.0732 0.0057 1.1345   0.1079 394.2440 0.0610 6.0967
## 8d      1.3963  0.3763 0.1300 0.9821   0.0917 338.1876 0.0605 6.0472
## 16     2.9260  0.7086 0.3780 0.7281   0.0665 390.4322 0.0469 4.6903   *
## 19b    -1.2633 -0.3232 0.1002 1.0305   0.0964 349.0457 0.0640 6.4005
## 23     -0.3663 -0.1091 0.0129 1.1428   0.1086 415.1281 0.0625 6.2461
## 26b    -1.1068 -0.2755 0.0753 1.0548   0.0996 431.9337 0.0590 5.9017
## 29b    -0.7368 -0.1148 0.0133 1.0401   0.1024 432.0087 0.0217 2.1662
## 35      1.7730  0.3141 0.0949 0.9527   0.0921 421.4731 0.0291 2.9123
## 36     -1.0965 -0.2851 0.0806 1.0602   0.0996 430.5517 0.0640 6.3972
## 39      0.0121 -0.0133 0.0002 1.1461   0.1093 415.2824 0.0602 6.0163
## 40      0.7874  0.1864 0.0354 1.0801   0.1028 409.6454 0.0555 5.5528

```

```

plot(inf) # plot influence diagnostics

```



```
# x-axis = each study's contribution to overall heterogeneity
# y-axis = its influence on the pooled estimate
# Build the study key from the model's data, in row order (Baujat labels by row position)
study_key <- paste0(
  seq_len(nrow(df_primary_logit)), # extract the number of rows from the dataframe
  " = ",
  df_primary_logit$Author) # author name would be combined with row number

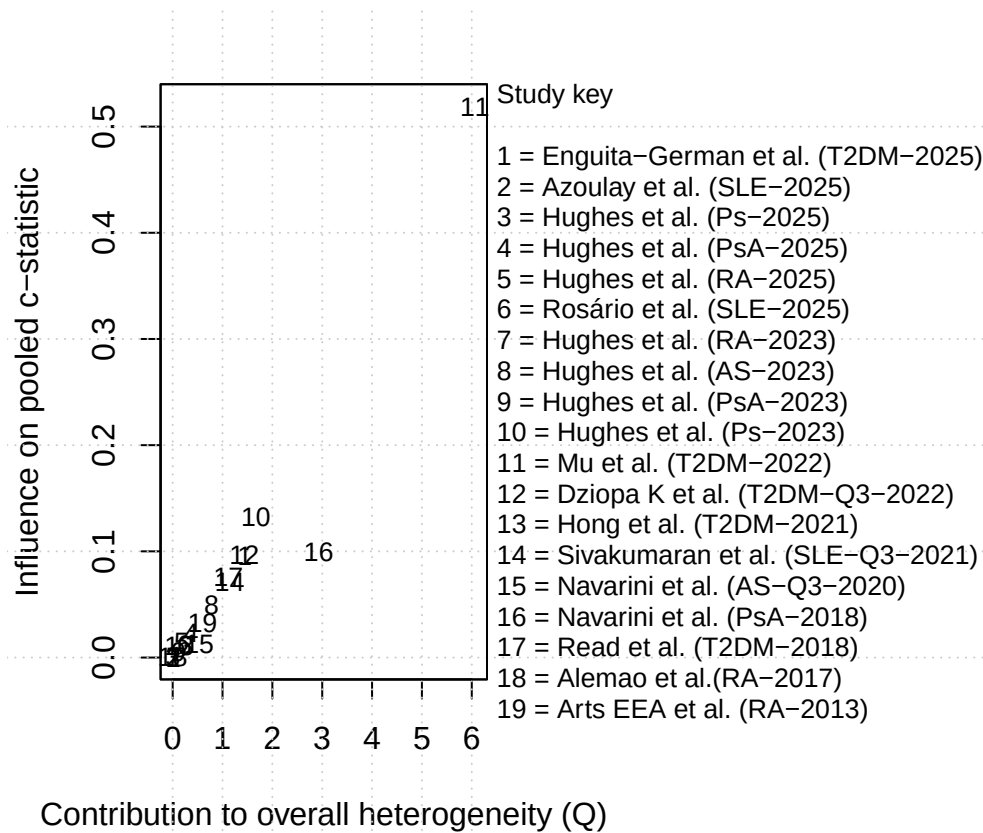
plot_settings <- par( # save the graphic setting inside
  mar = c(5, 4, 2, 20), # control lot margins to create better spacing
  xpd = NA) # allow text outside normal plotting region

baujat( # create baugat plot
  rma_meta,
  xlab = "Contribution to overall heterogeneity (Q)",
  ylab = "Influence on pooled c-statistic",
  main = ""
)

usr <- par("usr") # place study key outside the graph for identification

text(
  usr[2] + 0.2,
  usr[4],
  labels = paste(
    c("Study key", "", study_key),
```

```
collapse = "\n"),
adj = c(0, 1),
cex = 0.8
)
```



```
par(plot_settings)
```

```
# check difference in two dataframe
library(dplyr)
df_all_only <- anti_join( # keep only rows in first table that do not have a match in the
  ↪ second table
  df_all, # searching table
  df_primary, # reference table
  by = 'Author') # compare the two dataframe using the Author column
print(df_all_only) # print result
```

```
## # A tibble: 15 x 46
##   Number Index Author Title Sample_size Study_Year Disease_Group QRISK_Version
##   <dbl> <chr> <chr> <chr> <dbl> <dbl> <chr> <chr>
## 1     4 4 Krishn~ 04 C~ 5371 2024 "Diabetes (T~ QRISK3
## 2     5 5a Llorca~ 05 P~ 708 2012 "\r\n Rheuma~ QRISK3
## 3     5 5b Llorca~ 05 P~ 692 2012 "Ankylosing ~ QRISK3
## 4     5 5c Llorca~ 05 P~ 680 2012 "Psoriatic A~ QRISK3
## 5    14 14 Panopo~ 14 G~ 124 2023 "Systemic Lu~ QRISK3
```

```
## 6      18 18      Queved~ 18 {~          296      2022 "Systemic Lu~ QRISK3
## 7      19 19a     Dziopa~ 19 C~        168871     2022 "Diabetes (T~ QRISK2
## 8      21 21      Nedogo~ 21 I~          750      2021 "Metabolic S~ QRISK3
## 9      22 22      Carril~ 22 Q~         186      2021 "Inflammator~ QRISK3
## 10     26 26a     Sivaku~ 26 A~        1887      2021 "Systemic Lu~ QRISK2
## 11     27 27b     Drosos~ 27 U~         210      2020 "Systemic Lu~ QRISK3
## 12     28 28      Gerasi~ 28 A~         109      2021 "Rheumatoid ~ QRISK3
## 13     29 29a     Navari~ 29 C~         133      2020 "Ankylosing ~ QRISK2
## 14     32 32      Viswan~ 32 I~         339      2020 "chronic kid~ QRISK3
## 15     37 37      Shen e~ 37 U~         146      2018 "Psoriatic A~ QRISK2
## # i 38 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## #   `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## #   Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## #   Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## #   QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## #   Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## #   Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...
```

SUBGROUP ANALYSIS - Disease group

```
library(meta)

primary_subgroup_disease <- metagen(
  TE           = logit_c, # use logit estimates
  seTE         = se_logit, # use logit SE
  studlab      = Author, # identify by author
  data         = df_primary_logit, # find all in this dataframe
  sm           = "PLOGIT", # inform that the estimate are in logit
  method.tau   = "REML", # how to calculate the between study variance
  method.random.ci = "HK", # how to calcualte the CI
  subgroup     = Disease_Group, # split the studies according to disease group
  tau.common   = TRUE, # make all studeis in the group share the same tau square by
  ↪ pooling the estimates as few numbers of study would make seperate tau unstable
  prediction   = TRUE # calculate the prediction interval for each group
)

summary(primary_subgroup_disease)
```

##	proportion	95% CI	%W(common)
## Enguita-German et al. (T2DM-2025)	0.6630	[0.6453; 0.6803]	2.8
## Azoulay et al. (SLE-2025)	0.7600	[0.5001; 0.9093]	0.0
## Hughes et al. (Ps-2025)	0.7400	[0.7195; 0.7595]	1.6
## Hughes et al. (PsA-2025)	0.7000	[0.6361; 0.7569]	0.2
## Hughes et al. (RA-2025)	0.7160	[0.6904; 0.7403]	1.1
## Rosário et al. (SLE-2025)	0.7030	[0.5897; 0.7958]	0.1
## Hughes et al. (RA-2023)	0.7520	[0.7296; 0.7731]	1.3
## Hughes et al. (AS-2023)	0.7940	[0.7694; 0.8166]	0.8
## Hughes et al. (PsA-2023)	0.7640	[0.7379; 0.7883]	0.9
## Hughes et al. (Ps-2023)	0.8150	[0.7911; 0.8367]	0.8
## Mu et al. (T2DM-2022)	0.8780	[0.8317; 0.9129]	0.1
## Dziopa K et al. (T2DM-Q3-2022)	0.6640	[0.6600; 0.6680]	53.7
## Hong et al. (T2DM-2021)	0.7200	[0.6995; 0.7396]	1.7

```

## Sivakumaran et al. (SLE-Q3-2021)      0.6700 [0.6287; 0.7088]      0.5
## Navarini et al. (AS-Q3-2020)         0.6600 [0.4487; 0.8224]      0.0
## Navarini et al. (PsA-2018)           0.8660 [0.7660; 0.9273]      0.0
## Read et al. (T2DM-2018)              0.6740 [0.6690; 0.6790]     33.3
## Alemao et al.(RA-2017)               0.7440 [0.7128; 0.7729]      0.7
## Arts EEA et al. (RA-2013)           0.7900 [0.7468; 0.8276]      0.3
##                                     %W(random)      Disease_Group
## Enguita-German et al. (T2DM-2025)    6.3                      Diabetes (T2DM)
## Azoulay et al. (SLE-2025)            1.4 Systemic Lupus Erythematosus (SLE)
## Hughes et al. (Ps-2025)              6.2                      Psoriasis
## Hughes et al. (PsA-2025)             5.3                      Psoriatic Arthritis (PsA)
## Hughes et al. (RA-2025)              6.2                      Rheumatoid Arthritis (RA)
## Rosário et al. (SLE-2025)            3.9 Systemic Lupus Erythematosus (SLE)
## Hughes et al. (RA-2023)              6.2                      Rheumatoid Arthritis (RA)
## Hughes et al. (AS-2023)              6.1                      Ankylosing Spondylitis (AS)
## Hughes et al. (PsA-2023)            6.1                      Psoriatic Arthritis (PsA)
## Hughes et al. (Ps-2023)             6.0                      Psoriasis
## Mu et al. (T2DM-2022)               4.7                      Diabetes (T2DM)
## Dziopa K et al. (T2DM-Q3-2022)      6.4                      Diabetes (T2DM)
## Hong et al. (T2DM-2021)             6.2                      Diabetes (T2DM)
## Sivakumaran et al. (SLE-Q3-2021)    5.9 Systemic Lupus Erythematosus (SLE)
## Navarini et al. (AS-Q3-2020)        2.2                      Ankylosing Spondylitis (AS)
## Navarini et al. (PsA-2018)          2.9                      Psoriatic Arthritis (PsA)
## Read et al. (T2DM-2018)            6.4                      Diabetes (T2DM)
## Alemao et al.(RA-2017)             6.0                      Rheumatoid Arthritis (RA)
## Arts EEA et al. (RA-2013)          5.6                      Rheumatoid Arthritis (RA)
##
## Number of studies: k = 19
##
##                                     proportion      95% CI      z|t      p-value
## Common effect model                0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK)          0.7430 [0.7087; 0.7745] 12.94 < 0.0001
## Prediction interval                [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1166 [0.0518; 0.4064]; tau = 0.3414 [0.2277; 0.6375]
## I^2 = 89.2% [97.9%; 99.7%]; H = 3.04 [6.87; 19.05]
##
## Test of heterogeneity:
##      Q d.f.      p-value
## 432.04 18 < 0.0001
##
## Results for subgroups (common effect model):
##                                     k proportion
## Disease_Group = Diabetes (T2DM)    5      0.6691
## Disease_Group = Systemic Lupus Erythematosus (SLE) 3      0.6760
## Disease_Group = Psoriasis           2      0.7661
## Disease_Group = Psoriatic Arthritis (PsA) 3      0.7572
## Disease_Group = Rheumatoid Arthritis (RA) 4      0.7421
## Disease_Group = Ankylosing Spondylitis (AS) 2      0.7910

```

```

##                                     95% CI      Q    I2
## Disease_Group = Diabetes (T2DM)      [0.6661; 0.6721] 76.77 94.8%
## Disease_Group = Systemic Lupus Erythematosus (SLE) [0.6380; 0.7118] 0.84 0.0%
## Disease_Group = Psoriasis             [0.7504; 0.7811] 21.74 95.4%
## Disease_Group = Psoriatic Arthritis (PsA) [0.7337; 0.7792] 8.56 76.6%
## Disease_Group = Rheumatoid Arthritis (RA) [0.7282; 0.7555] 9.89 69.7%
## Disease_Group = Ankylosing Spondylitis (AS) [0.7665; 0.8135] 2.33 57.1%
##
## Test for subgroup differences (common effect model):
##           Q d.f.  p-value
## Between groups 311.92    5 < 0.0001
## Within groups  120.12   13 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                                     k proportion
## Disease_Group = Diabetes (T2DM)      5      0.7216
## Disease_Group = Systemic Lupus Erythematosus (SLE) 3      0.6950
## Disease_Group = Psoriasis             2      0.7793
## Disease_Group = Psoriatic Arthritis (PsA) 3      0.7690
## Disease_Group = Rheumatoid Arthritis (RA) 4      0.7506
## Disease_Group = Ankylosing Spondylitis (AS) 2      0.7608
##                                     95% CI  tau2
## Disease_Group = Diabetes (T2DM)      [0.5784; 0.8304] 0.1166
## Disease_Group = Systemic Lupus Erythematosus (SLE) [0.5914; 0.7820] 0.1166
## Disease_Group = Psoriasis             [0.1804; 0.9827] 0.1166
## Disease_Group = Psoriatic Arthritis (PsA) [0.5149; 0.9126] 0.1166
## Disease_Group = Rheumatoid Arthritis (RA) [0.6988; 0.7961] 0.1166
## Disease_Group = Ankylosing Spondylitis (AS) [0.0597; 0.9938] 0.1166
##                                     tau
## Disease_Group = Diabetes (T2DM)      0.3414
## Disease_Group = Systemic Lupus Erythematosus (SLE) 0.3414
## Disease_Group = Psoriasis             0.3414
## Disease_Group = Psoriatic Arthritis (PsA) 0.3414
## Disease_Group = Rheumatoid Arthritis (RA) 0.3414
## Disease_Group = Ankylosing Spondylitis (AS) 0.3414
##
## Test for subgroup differences (random effects model (HK)):
##           Q d.f.  p-value
## Between groups  6.43    5  0.2665
## Within groups  120.12   13 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau2
##   (assuming common tau2 in subgroups)
## - Q-Profile method for confidence interval of tau2 and tau
## - Calculation of I2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation

```

FORREST PLOT - SUBGROUP ANALYSIS - Disease group

```
meta::forest(
  primary_subgroup_disease,
  backtransf = TRUE,
  xlab = "c-statistic",
  xlim = c(0.5, 1),
  at = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
  leftcols = c("studlab", "Sample_size", "Events"),
  leftlabs = c("Study", "N", "CVD events"),
  rightcols = c("effect", "ci"),
  rightlabs = c("c-statistic", "95% CI"),
  common = FALSE,
  test.subgroup = TRUE,
  print.subgroup.name = FALSE,
  addrows.below.overall = 5,
  colgap = "8mm",
  colgap.left = "6mm",
  colgap.forest.left = "10mm",
  fontsize = 11
)
```

0%, $\tau^2 = 0.1166$, $p = 0.6584$

2025)	8062	607
2023)	390664	19078

Model (HK)

5.4%, $\tau^2 = 0.1166$, $p < 0.0001$

(PsA)

-2025)	769	70
-2023)	26375	1802
-2018)	155	21

Model (HK)

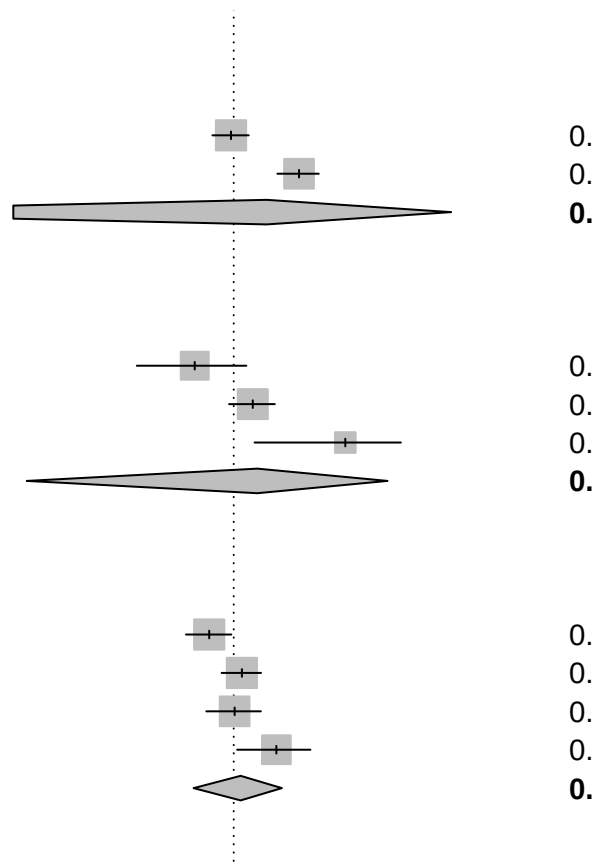
6.6%, $\tau^2 = 0.1166$, $p = 0.0139$

ritis (RA)

2025)	4772	485
2023)	91750	10569
2017)	12747	1361
-2013)	1050	126

Model (HK)

9.7%, $\tau^2 = 0.1166$, $p = 0.0195$



```
# Between-group difference test
primary_subgroup_disease$Q.b.random      # Q between random effect
```

```
## [1] 6.430947
```



```
primary_subgroup_disease$df.Q.b           # degree of freedom
```

```
## [1] 5
```

```
primary_subgroup_disease$pval.Q.b.random # p-value
```

```
## [1] 0.2665139
```

```
# Per-subgroup back-transformed estimates
data.frame(
  disease = primary_subgroup_disease$subgroup.levels,
  k       = primary_subgroup_disease$k.w,
  c_stat  = plogis(primary_subgroup_disease$TE.random.w),
  lower   = plogis(primary_subgroup_disease$lower.random.w),
  upper   = plogis(primary_subgroup_disease$upper.random.w),
  I2      = primary_subgroup_disease$I2.w
)
```

```
##                                     disease k
## Diabetes (T2DM)                    Diabetes (T2DM) 5
## Systemic Lupus Erythematosus (SLE) Systemic Lupus Erythematosus (SLE) 3
## Psoriasis                          Psoriasis 2
## Psoriatic Arthritis (PsA)          Psoriatic Arthritis (PsA) 3
## Rheumatoid Arthritis (RA)          Rheumatoid Arthritis (RA) 4
## Ankylosing Spondylitis (AS)        Ankylosing Spondylitis (AS) 2
##                                     c_stat   lower   upper   I2
## Diabetes (T2DM)                    0.7215913 0.5783504 0.8304376 0.9478949
## Systemic Lupus Erythematosus (SLE) 0.6949905 0.5913864 0.7820087 0.0000000
## Psoriasis                          0.7792959 0.1804002 0.9826519 0.9539983
## Psoriatic Arthritis (PsA)          0.7689805 0.5149041 0.9125753 0.7663248
## Rheumatoid Arthritis (RA)          0.7505931 0.6988173 0.7960656 0.6965992
## Ankylosing Spondylitis (AS)        0.7607797 0.0596687 0.9937651 0.5705511
```

SUBGROUP ANALYSIS BY QRISK VERSION

```
# See version information
table(df_primary_logit$QRISK_Version, useNA = "ifany")
```

```
##
## QRISK2 QRISK3
##      5      14
```

```
# subgroup analysis QRISK version
library(meta)

primary_subgroup_version <- metagen(
  TE      = logit_c,
  seTE    = se_logit,
```

```

studlab      = Author,
data         = df_primary_logit,
sm           = "PLOGIT",
method.tau   = "REML",
method.random.ci = "HK",
subgroup     = QRISK_Version,
tau.common   = TRUE,
prediction   = TRUE
)

summary(primary_subgroup_version)

```

	proportion	95% CI	%W(common)
## Enguita-German et al. (T2DM-2025)	0.6630	[0.6453; 0.6803]	2.8
## Azoulay et al. (SLE-2025)	0.7600	[0.5001; 0.9093]	0.0
## Hughes et al. (Ps-2025)	0.7400	[0.7195; 0.7595]	1.6
## Hughes et al. (PsA-2025)	0.7000	[0.6361; 0.7569]	0.2
## Hughes et al. (RA-2025)	0.7160	[0.6904; 0.7403]	1.1
## Rosário et al. (SLE-2025)	0.7030	[0.5897; 0.7958]	0.1
## Hughes et al. (RA-2023)	0.7520	[0.7296; 0.7731]	1.3
## Hughes et al. (AS-2023)	0.7940	[0.7694; 0.8166]	0.8
## Hughes et al. (PsA-2023)	0.7640	[0.7379; 0.7883]	0.9
## Hughes et al. (Ps-2023)	0.8150	[0.7911; 0.8367]	0.8
## Mu et al. (T2DM-2022)	0.8780	[0.8317; 0.9129]	0.1
## Dziopa K et al. (T2DM-Q3-2022)	0.6640	[0.6600; 0.6680]	53.7
## Hong et al. (T2DM-2021)	0.7200	[0.6995; 0.7396]	1.7
## Sivakumaran et al. (SLE-Q3-2021)	0.6700	[0.6287; 0.7088]	0.5
## Navarini et al. (AS-Q3-2020)	0.6600	[0.4487; 0.8224]	0.0
## Navarini et al. (PsA-2018)	0.8660	[0.7660; 0.9273]	0.0
## Read et al. (T2DM-2018)	0.6740	[0.6690; 0.6790]	33.3
## Alemao et al. (RA-2017)	0.7440	[0.7128; 0.7729]	0.7
## Arts EEA et al. (RA-2013)	0.7900	[0.7468; 0.8276]	0.3
##	%W(random)	QRISK_Version	
## Enguita-German et al. (T2DM-2025)	6.3	QRISK2	
## Azoulay et al. (SLE-2025)	1.4	QRISK3	
## Hughes et al. (Ps-2025)	6.2	QRISK3	
## Hughes et al. (PsA-2025)	5.3	QRISK3	
## Hughes et al. (RA-2025)	6.2	QRISK3	
## Rosário et al. (SLE-2025)	3.9	QRISK3	
## Hughes et al. (RA-2023)	6.2	QRISK3	
## Hughes et al. (AS-2023)	6.1	QRISK3	
## Hughes et al. (PsA-2023)	6.1	QRISK3	
## Hughes et al. (Ps-2023)	6.0	QRISK3	
## Mu et al. (T2DM-2022)	4.7	QRISK3	
## Dziopa K et al. (T2DM-Q3-2022)	6.4	QRISK3	
## Hong et al. (T2DM-2021)	6.2	QRISK3	
## Sivakumaran et al. (SLE-Q3-2021)	5.9	QRISK3	
## Navarini et al. (AS-Q3-2020)	2.2	QRISK3	
## Navarini et al. (PsA-2018)	2.9	QRISK2	
## Read et al. (T2DM-2018)	6.4	QRISK2	
## Alemao et al. (RA-2017)	6.0	QRISK2	
## Arts EEA et al. (RA-2013)	5.6	QRISK2	
##			

```

## Number of studies: k = 19
##
##
##           proportion          95% CI    z|t  p-value
## Common effect model          0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK)    0.7430 [0.7087; 0.7745]  12.94 < 0.0001
## Prediction interval          [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1075 [0.0529; 0.2999]; tau = 0.3279 [0.2300; 0.5477]
## I^2 = 96.1% [94.3%; 98.9%]; H = 5.04 [4.19; 9.73]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 432.04   18 < 0.0001
##
## Results for subgroups (common effect model):
##      k proportion          95% CI      Q  I^2
## QRISK_Version = QRISK2    5      0.6758 [0.6711; 0.6806]  52.77 92.4%
## QRISK_Version = QRISK3   14      0.6769 [0.6733; 0.6806] 379.13 96.6%
##
## Test for subgroup differences (common effect model):
##      Q d.f.  p-value
## Between groups    0.13    1    0.7188
## Within groups  431.91   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##      k proportion          95% CI  tau^2   tau
## QRISK_Version = QRISK2    5      0.7385 [0.6245; 0.8275] 0.1075 0.3279
## QRISK_Version = QRISK3   14      0.7449 [0.7048; 0.7813] 0.1075 0.3279
##
## Test for subgroup differences (random effects model (HK)):
##      Q d.f.  p-value
## Between groups    0.02    1    0.8751
## Within groups  431.91   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation

# Extract the between-group test and per-version estimate
# Between-group difference
primary_subgroup_version$Q.b.random

## [1] 0.02471093

```

```
primary_subgroup_version$df.Q.b
```

```
## [1] 1
```

```
primary_subgroup_version$pval.Q.b.random
```

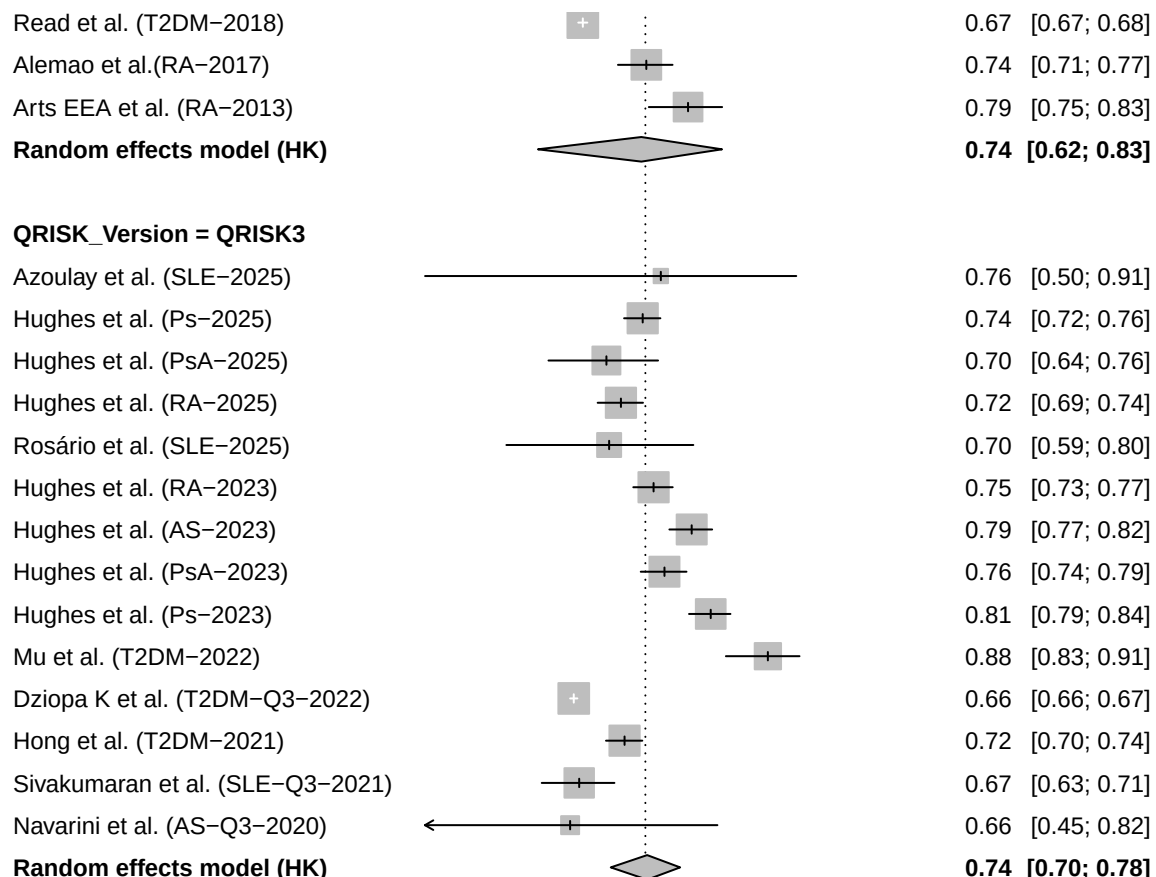
```
## [1] 0.8750895
```

```
# Per-version back-transformed estimates
data.frame(
  version = primary_subgroup_version$subgroup.levels,
  k       = primary_subgroup_version$k.w,
  c_stat  = plogis(primary_subgroup_version$TE.random.w),
  lower   = plogis(primary_subgroup_version$lower.random.w),
  upper   = plogis(primary_subgroup_version$upper.random.w),
  I2      = primary_subgroup_version$I2.w
)
```

```
##      version k    c_stat    lower    upper      I2
## QRISK2  QRISK2  5 0.7385233 0.6245188 0.8274757 0.9242035
## QRISK3  QRISK3 14 0.7449155 0.7048192 0.7812557 0.9657114
```

```
# widen the bottom margin before plotting
par(mar = c(8, 4, 2, 2)) # bottom, left, top, right - more bottom space
```

```
meta::forest(primary_subgroup_version,
  backtransf = TRUE,
  xlab       = "c-statistic",
  xlim      = c(0.5, 1),
  at        = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
  leftcols  = "studlab", leftlabs = "Study",
  rightcols = c("effect", "ci"), rightlabs = c("c-statistic", "95% CI"),
  common    = FALSE,
  hetstat   = FALSE,
  test.subgroup = TRUE,
  overall.hetstat = TRUE,
  fontsize  = 9,
  spacing   = 1.1)
```

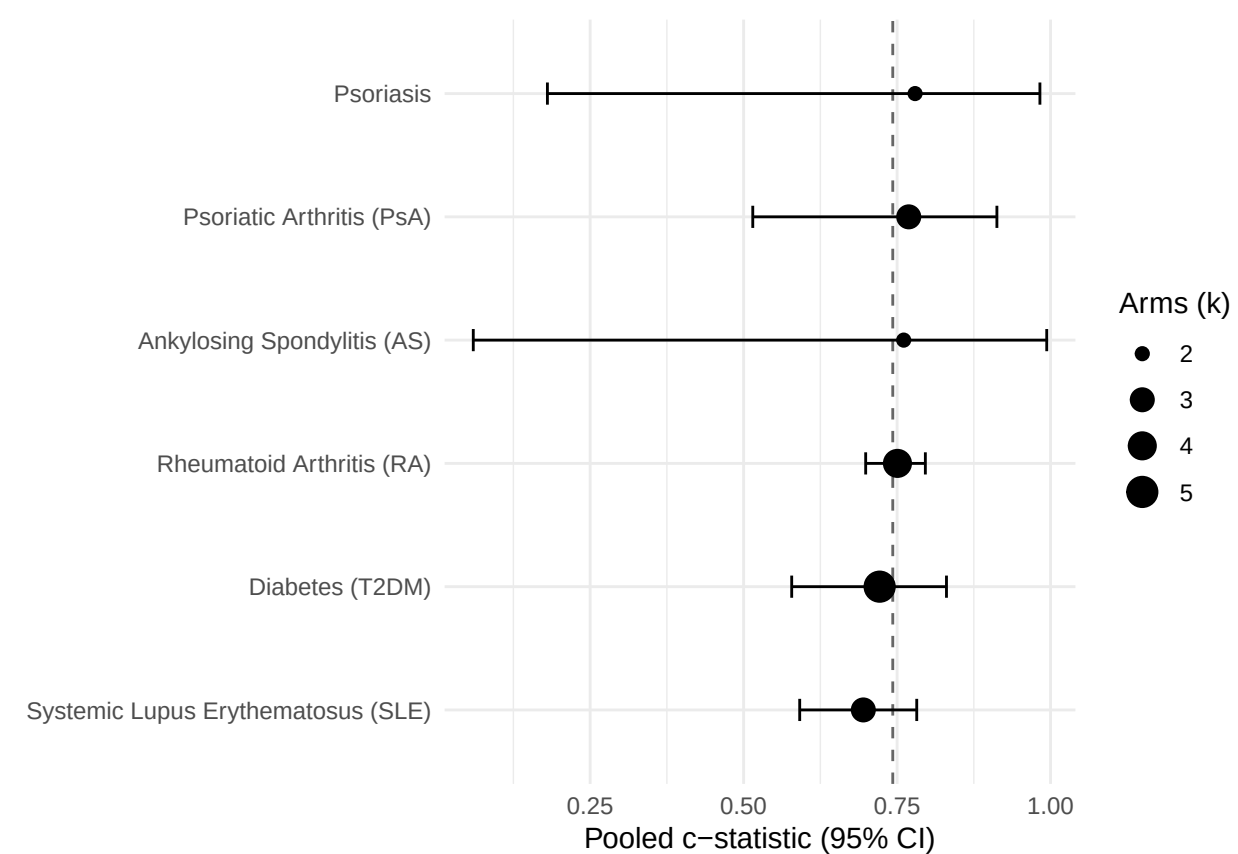


```
# subgroup summary
library(ggplot2)
sg <- data.frame(
  group = primary_subgroup_disease$subgroup.levels,
  k      = primary_subgroup_disease$k.w,
  est    = plogis(primary_subgroup_disease$TE.random.w),
  lo     = plogis(primary_subgroup_disease$lower.random.w),
  hi     = plogis(primary_subgroup_disease$upper.random.w))

ggplot(sg, aes(x = est, y = reorder(group, est))) +
  geom_vline(xintercept = plogis(primary_logit$TE.random),
             linetype = 2, colour = "grey40") +
  geom_errorbarh(aes(xmin = lo, xmax = hi), height = 0.18) +
  geom_point(aes(size = k)) +
  scale_size_continuous(range = c(2, 5), name = "Arms (k)") +
  labs(x = "Pooled c-statistic (95% CI)", y = NULL) +
  theme_minimal(base_size = 11)
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `height` was translated to `width`.
```



```
primary_subgroup_disease$Q.b.random
```

```
## [1] 6.430947
```

```
primary_subgroup_disease$df.Q.b
```

```
## [1] 5
```

```
primary_subgroup_disease$pval.Q.b.random
```

```
## [1] 0.2665139
```

PAIRED QRISK VERSION COMPARISM

```
library(dplyr)
library(tidyr)
```

```
##
## Attaching package: 'tidyr'
```

```
## The following objects are masked from 'package:Matrix':
##
##     expand, pack, unpack
```

```
library(readr)

# Build the paired set
paired <- df_all %>%
  filter(Index %in% c("19a", "19b", "26a", "26b", "29a", "29b")) %>%
  mutate(
    Index = trimws(Index), # trim whitespace from the index
    cohort = parse_number(Index), # extract the number part from the index
    version = QRISK_Version # store the qrisk version in version
  )

# verify 6 rows, each cohort has one QRISK2 and one QRISK3
print(count(paired, cohort, version))
```

```
## # A tibble: 6 x 3
##   cohort version      n
##   <dbl> <chr>   <int>
## 1     19 QRISK2      1
## 2     19 QRISK3      1
## 3     26 QRISK2      1
## 4     26 QRISK3      1
## 5     29 QRISK2      1
## 6     29 QRISK3      1
```

```
# Logit-transform each arm c-statistic
paired <- paired %>%
  mutate( # create column for each new logit transformed variable
    logit_c = log(Estimate / (1 - Estimate)),
    logit_lb = log(LowerCI / (1 - LowerCI)),
    logit_ub = log(UpperCI / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96)
  )

# Reshape to one row per cohort, QRISK2 and QRISK3 side by side
wide <- paired %>%
  select(cohort, version, Estimate, logit_c, se_logit) %>%
  pivot_wider( # convert to data to wide form
    id_cols = cohort,
    names_from = version,
    values_from = c(Estimate, logit_c, se_logit) # variables to be widened
  )

print(names(wide))
```

```
## [1] "cohort"      "Estimate_QRISK2" "Estimate_QRISK3" "logit_c_QRISK2"
## [5] "logit_c_QRISK3" "se_logit_QRISK2" "se_logit_QRISK3"
```

```

# Within-cohort difference (QRISK3 - QRISK2)
wide <- wide %>%
  mutate(
    delta_logit    = logit_c_QRISK3 - logit_c_QRISK2,
    delta_cstat    = Estimate_QRISK3 - Estimate_QRISK2,
    se_delta_indep = sqrt(se_logit_QRISK3^2 + se_logit_QRISK2^2) # SE assuming
    ↪ independence
  )

# Present the paired comparison
wide %>%
  transmute(
    Cohort          = cohort,
    `QRISK2 c`      = round(Estimate_QRISK2, 3),
    `QRISK3 c`      = round(Estimate_QRISK3, 3),
    `Delta c (3-2)` = round(delta_cstat, 3),
    `Delta logit (3-2)` = round(delta_logit, 3),
    `SE (indep, overest)` = round(se_delta_indep, 3)
  )

```

```

## # A tibble: 3 x 6
##   Cohort `QRISK2 c` `QRISK3 c` `Delta c (3-2)` `Delta logit (3-2)`
##   <dbl>   <dbl>   <dbl>         <dbl>         <dbl>
## 1     19     0.664     0.664           0           0
## 2     26     0.72     0.67        -0.05       -0.236
## 3     29     0.68     0.66        -0.02       -0.09
## # i 1 more variable: `SE (indep, overest)` <dbl>

```

```

# check the average improvement of discrimination between 3 and 2
mean_delta_logit <- mean(wide$delta_logit)
cat("Mean within-cohort logit difference (QRISK3 - QRISK2):",
    round(mean_delta_logit, 4), "\n")

```

```

## Mean within-cohort logit difference (QRISK3 - QRISK2): -0.1089

```

SUBGROUP ANALYSIS (PROBAST)

```

# Which column holds the overall PROBAST ROB verdict, and how is it coded?
table(df_primary_logit$Overall, useNA = "ifany")

```

```

##
## HIGH  LOW
##    11    8

```

```

# ROB SUBGROUP ANALYSIS
library(meta)

primary_subgroup_rob <- metagen(
  TE          = logit_c,

```



```

seTE          = se_logit,
studlab       = Author,
data          = df_primary_logit,
sm            = "PLOGIT",
method.tau    = "REML",
method.random.ci = "HK",
subgroup      = Overall,
tau.common    = TRUE,
prediction    = TRUE
)

summary(primary_subgroup_rob)

```

	proportion	95% CI	%W(common)
## Enguita-German et al. (T2DM-2025)	0.6630	[0.6453; 0.6803]	2.8
## Azoulay et al. (SLE-2025)	0.7600	[0.5001; 0.9093]	0.0
## Hughes et al. (Ps-2025)	0.7400	[0.7195; 0.7595]	1.6
## Hughes et al. (PsA-2025)	0.7000	[0.6361; 0.7569]	0.2
## Hughes et al. (RA-2025)	0.7160	[0.6904; 0.7403]	1.1
## Rosário et al. (SLE-2025)	0.7030	[0.5897; 0.7958]	0.1
## Hughes et al. (RA-2023)	0.7520	[0.7296; 0.7731]	1.3
## Hughes et al. (AS-2023)	0.7940	[0.7694; 0.8166]	0.8
## Hughes et al. (PsA-2023)	0.7640	[0.7379; 0.7883]	0.9
## Hughes et al. (Ps-2023)	0.8150	[0.7911; 0.8367]	0.8
## Mu et al. (T2DM-2022)	0.8780	[0.8317; 0.9129]	0.1
## Dziopa K et al. (T2DM-Q3-2022)	0.6640	[0.6600; 0.6680]	53.7
## Hong et al. (T2DM-2021)	0.7200	[0.6995; 0.7396]	1.7
## Sivakumaran et al. (SLE-Q3-2021)	0.6700	[0.6287; 0.7088]	0.5
## Navarini et al. (AS-Q3-2020)	0.6600	[0.4487; 0.8224]	0.0
## Navarini et al. (PsA-2018)	0.8660	[0.7660; 0.9273]	0.0
## Read et al. (T2DM-2018)	0.6740	[0.6690; 0.6790]	33.3
## Alemao et al. (RA-2017)	0.7440	[0.7128; 0.7729]	0.7
## Arts EEA et al. (RA-2013)	0.7900	[0.7468; 0.8276]	0.3
##	%W(random)	Overall	
## Enguita-German et al. (T2DM-2025)	6.3	HIGH	
## Azoulay et al. (SLE-2025)	1.4	HIGH	
## Hughes et al. (Ps-2025)	6.2	LOW	
## Hughes et al. (PsA-2025)	5.3	LOW	
## Hughes et al. (RA-2025)	6.2	LOW	
## Rosário et al. (SLE-2025)	3.9	HIGH	
## Hughes et al. (RA-2023)	6.2	LOW	
## Hughes et al. (AS-2023)	6.1	LOW	
## Hughes et al. (PsA-2023)	6.1	LOW	
## Hughes et al. (Ps-2023)	6.0	LOW	
## Mu et al. (T2DM-2022)	4.7	HIGH	
## Dziopa K et al. (T2DM-Q3-2022)	6.4	LOW	
## Hong et al. (T2DM-2021)	6.2	HIGH	
## Sivakumaran et al. (SLE-Q3-2021)	5.9	HIGH	
## Navarini et al. (AS-Q3-2020)	2.2	HIGH	
## Navarini et al. (PsA-2018)	2.9	HIGH	
## Read et al. (T2DM-2018)	6.4	HIGH	
## Alemao et al. (RA-2017)	6.0	HIGH	
## Arts EEA et al. (RA-2013)	5.6	HIGH	

```

##
## Number of studies: k = 19
##
##               proportion           95% CI      z|t  p-value
## Common effect model           0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK)      0.7430 [0.7087; 0.7745] 12.94 < 0.0001
## Prediction interval           [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1079 [0.0530; 0.3000]; tau = 0.3285 [0.2303; 0.5478]
## I^2 = 96.1% [94.3%; 98.9%]; H = 5.03 [4.19; 9.73]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 432.04   18 < 0.0001
##
## Results for subgroups (common effect model):
##      k proportion           95% CI      Q  I^2
## Overall = HIGH  11      0.6787 [0.6741; 0.6832] 111.02 91.0%
## Overall = LOW   8       0.6751 [0.6714; 0.6788] 319.58 97.8%
##
## Test for subgroup differences (common effect model):
##      Q d.f.  p-value
## Between groups  1.43   1  0.2313
## Within groups  430.60  17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##      k proportion           95% CI  tau^2   tau
## Overall = HIGH  11      0.7405 [0.6793; 0.7936] 0.1079 0.3285
## Overall = LOW   8       0.7460 [0.7021; 0.7854] 0.1079 0.3285
##
## Test for subgroup differences (random effects model (HK)):
##      Q d.f.  p-value
## Between groups  0.03   1  0.8594
## Within groups  430.60  17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation

# Between-group test and per-group estimates
# Between-group difference
primary_subgroup_rob$Q.b.random

```

```
## [1] 0.03135599
```

```
primary_subgroup_rob$df.Q.b
```

```
## [1] 1
```

```
primary_subgroup_rob$pval.Q.b.random
```

```
## [1] 0.8594485
```

```
# Per-ROB-group back-transformed estimates
data.frame(
  rob      = primary_subgroup_rob$subgroup.levels,
  k        = primary_subgroup_rob$k.w,
  c_stat   = plogis(primary_subgroup_rob$TE.random.w),
  lower    = plogis(primary_subgroup_rob$lower.random.w),
  upper    = plogis(primary_subgroup_rob$upper.random.w),
  I2       = primary_subgroup_rob$I2.w
)
```

```
##      rob k    c_stat    lower    upper    I2
## HIGH HIGH 11 0.7405318 0.6793419 0.7935928 0.9099277
## LOW  LOW  8 0.7460371 0.7021353 0.7854456 0.9780964
```

```
# FORST PLOT FOR ROB
par(mar = c(8, 4, 2, 2))

meta::forest(
  primary_subgroup_rob,
  backtransf = TRUE,
  xlab = "c-statistic",
  xlim = c(0.5, 1),
  at = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),

  leftcols = "studlab",
  leftlabs = "Study",

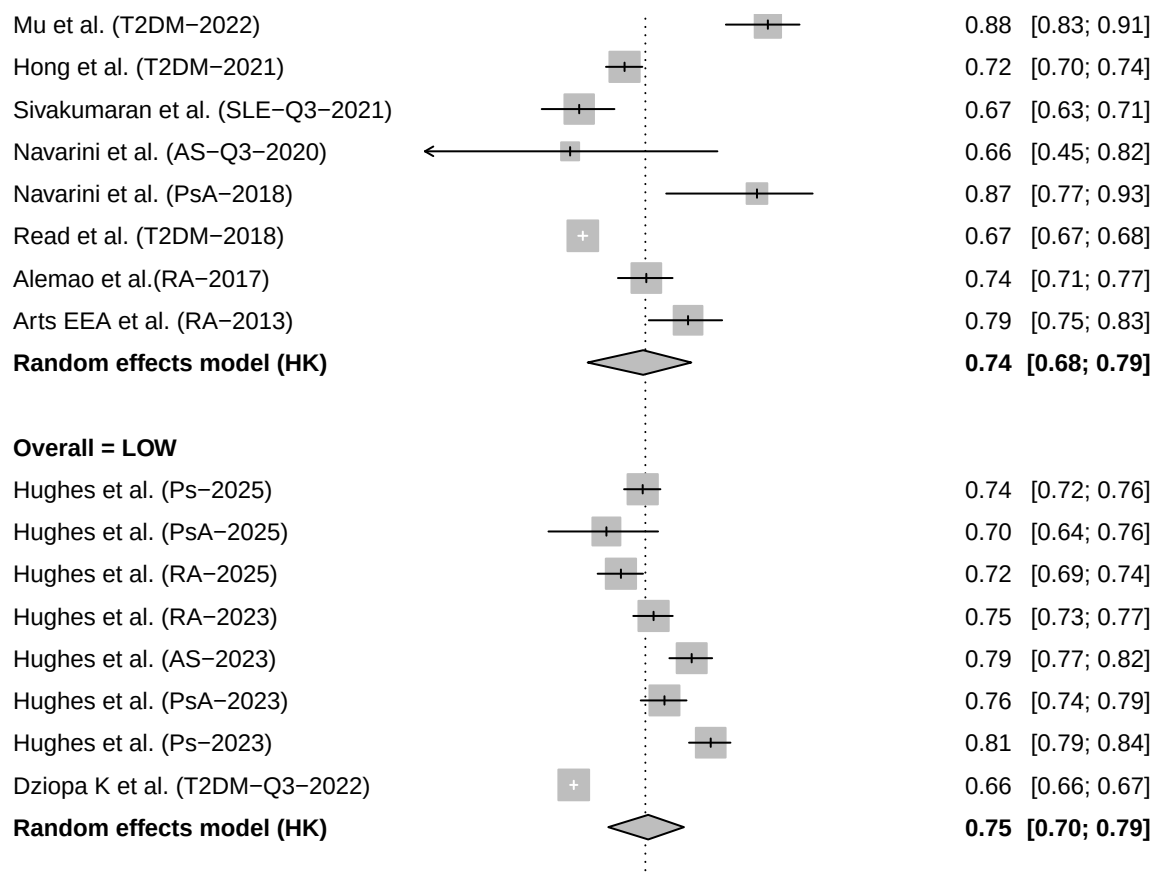
  rightcols = c("effect", "ci"),
  rightlabs = c("c-statistic", "95% CI"),

  common = FALSE,

  hetstat = FALSE,
  overall.hetstat = TRUE,
  test.subgroup = TRUE,

  fontsize = 9,
  spacing = 1.1,

  addrows.below.overall = 2
)
```



SENSITIVITY ANALYSIS DROPPING LOW STUDIES

```
df_lowrob <- df_primary_logit[df_primary_logit$Overall == "LOW", ]

primary_lowrob <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = df_lowrob,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  prediction = TRUE
)
summary(primary_lowrob)
```

##	proportion	95% CI	%W(common)	%W(random)
## 6a	0.7400	[0.7195; 0.7595]	2.6	13.0
## 6b	0.7000	[0.6361; 0.7569]	0.3	10.2
## 6c	0.7160	[0.6904; 0.7403]	1.9	12.8
## 8a	0.7520	[0.7296; 0.7731]	2.1	12.9
## 8b	0.7940	[0.7694; 0.8166]	1.4	12.5
## 8c	0.7640	[0.7379; 0.7883]	1.5	12.6

```
## 8d      0.8150 [0.7911; 0.8367]      1.2      12.4
## 19b     0.6640 [0.6600; 0.6680]     88.9      13.6
##
## Number of studies: k = 8
##
##              proportion          95% CI    z|t    p-value
## Common effect model      0.6751 [0.6714; 0.6788] 84.78      0
## Random effects model (HK) 0.7460 [0.7020; 0.7855] 11.56 < 0.0001
## Prediction interval      [0.6072; 0.8481]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0649 [0.0256; 0.2804]; tau = 0.2548 [0.1599; 0.5295]
## I^2 = 97.8% [96.9%; 98.4%]; H = 6.76 [5.68; 8.03]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 319.58    7 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 7)
## - Prediction interval based on t-distribution (df = 7)
## - Logit transformation
```

SENSITIVITY ANALYSIS FOR DISCRIMINATION METRIC TYPE

```
# CHECK TYPE
table(df_primary_logit$Dscrimination, useNA = "ifany")

##
##              AUC Harrell's C-index
##              15                  4

# LEAVE OUT ONLY HARRELLS
library(meta)

# Keep only the AUC-metric studies (drop the three Harrell's C arms)
df_auc_only <- df_primary_logit[df_primary_logit$Dscrimination == "AUC", ]
nrow(df_auc_only)

## [1] 15

sens_auc_only <- metagen(
  TE      = logit_c,
  seTE    = se_logit,
  studlab = Index,
  data    = df_auc_only,
```

```

sm          = "PLOGIT",
method.tau  = "REML",
method.random.ci = "HK",
prediction  = TRUE
)
summary(sens_auc_only)

```

```

##      proportion      95% CI %W(common) %W(random)
## 3      0.7600 [0.5001; 0.9093]      0.1      1.8
## 6a     0.7400 [0.7195; 0.7595]     16.8      8.4
## 6b     0.7000 [0.6361; 0.7569]      2.2      7.0
## 6c     0.7160 [0.6904; 0.7403]     12.0      8.3
## 7      0.7030 [0.5897; 0.7958]      0.7      5.1
## 8a     0.7520 [0.7296; 0.7731]     13.3      8.4
## 8b     0.7940 [0.7694; 0.8166]      8.7      8.2
## 8c     0.7640 [0.7379; 0.7883]      9.3      8.2
## 8d     0.8150 [0.7911; 0.8367]      7.9      8.1
## 16     0.8780 [0.8317; 0.9129]      1.3      6.2
## 23     0.7200 [0.6995; 0.7396]     18.4      8.4
## 26b    0.6700 [0.6287; 0.7088]      5.5      7.9
## 29b    0.6600 [0.4487; 0.8224]      0.2      2.7
## 35     0.8660 [0.7660; 0.9273]      0.4      3.7
## 40     0.7900 [0.7468; 0.8276]      3.1      7.4
##
## Number of studies: k = 15
##
##      proportion      95% CI  z|t  p-value
## Common effect model      0.7489 [0.7408; 0.7569] 50.27      0
## Random effects model (HK) 0.7602 [0.7226; 0.7941] 12.61 < 0.0001
## Prediction interval      [0.6178; 0.8614]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0907 [0.0387; 0.3101]; tau = 0.3012 [0.1967; 0.5568]
## I^2 = 86.9% [80.0%; 91.4%]; H = 2.76 [2.24; 3.41]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 106.96  14 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 14)
## - Prediction interval based on t-distribution (df = 14)
## - Logit transformation

```

SUBGROUP ANALYSIS UK vs Non UK

```
table(df_primary_logit$Country, useNA = "ifany")
```

```
##
##      Brazil      Canada      China      France      Italy Netherlands
##      1          1          1          1          2          1
##      Spain      UK          USA
##      1          10         1
```

```
library(dplyr)
library(meta)

# Collapse to UK vs non-UK
df_primary_logit <- df_primary_logit %>%
  mutate(UK_group = ifelse(Country == "UK", "UK", "Non-UK"))

table(df_primary_logit$UK_group)
```

```
##
## Non-UK      UK
##      9      10
```

```
primary_subgroup_uk <- metagen(
  TE      = logit_c,
  seTE     = se_logit,
  studlab  = Index,
  data     = df_primary_logit,
  sm       = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  subgroup  = UK_group,
  tau.common = TRUE,
  prediction = TRUE
)
summary(primary_subgroup_uk)
```

```
##      proportion      95% CI %W(common) %W(random) UK_group
## 1      0.6630 [0.6453; 0.6803]      2.8      6.3      Non-UK
## 3      0.7600 [0.5001; 0.9093]      0.0      1.4      Non-UK
## 6a     0.7400 [0.7195; 0.7595]      1.6      6.2        UK
## 6b     0.7000 [0.6361; 0.7569]      0.2      5.3        UK
## 6c     0.7160 [0.6904; 0.7403]      1.1      6.2        UK
## 7      0.7030 [0.5897; 0.7958]      0.1      3.9      Non-UK
## 8a     0.7520 [0.7296; 0.7731]      1.3      6.2        UK
## 8b     0.7940 [0.7694; 0.8166]      0.8      6.1        UK
## 8c     0.7640 [0.7379; 0.7883]      0.9      6.1        UK
## 8d     0.8150 [0.7911; 0.8367]      0.8      6.0        UK
## 16     0.8780 [0.8317; 0.9129]      0.1      4.7      Non-UK
## 19b    0.6640 [0.6600; 0.6680]     53.7      6.4        UK
## 23     0.7200 [0.6995; 0.7396]      1.7      6.2      Non-UK
## 26b    0.6700 [0.6287; 0.7088]      0.5      5.9      Non-UK
## 29b    0.6600 [0.4487; 0.8224]      0.0      2.2      Non-UK
```

```

## 35      0.8660 [0.7660; 0.9273]      0.0      2.9  Non-UK
## 36      0.6740 [0.6690; 0.6790]     33.3      6.4      UK
## 39      0.7440 [0.7128; 0.7729]      0.7      6.0      UK
## 40      0.7900 [0.7468; 0.8276]      0.3      5.6  Non-UK
##
## Number of studies: k = 19
##
##              proportion      95% CI      z|t  p-value
## Common effect model      0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK)  0.7430 [0.7087; 0.7745] 12.94 < 0.0001
## Prediction interval      [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1087 [0.0534; 0.2961]; tau = 0.3297 [0.2311; 0.5441]
## I^2 = 95.9% [97.5%; 99.5%]; H = 4.97 [6.32; 14.72]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 432.04   18 < 0.0001
##
## Results for subgroups (common effect model):
##      k proportion      95% CI      Q  I^2
## UK_group = Non-UK    9      0.6975 [0.6857; 0.7091] 82.29 90.3%
## UK_group = UK       10      0.6753 [0.6723; 0.6782] 337.07 97.3%
##
## Test for subgroup differences (common effect model):
##      Q d.f.  p-value
## Between groups 12.68   1  0.0004
## Within groups 419.36  17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##      k proportion      95% CI  tau^2  tau
## UK_group = Non-UK    9      0.7499 [0.6725; 0.8141] 0.1087 0.3297
## UK_group = UK       10      0.7388 [0.7017; 0.7728] 0.1087 0.3297
##
## Test for subgroup differences (random effects model (HK)):
##      Q d.f.  p-value
## Between groups  0.10   1  0.7500
## Within groups 419.36  17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation

```


SUGGROUP ANALYSIS - FOLLOW UP

```
# See the follow-up values across the pool
df_primary_logit %>%
  select(Index, Disease_Group, `Follow-Up`, Estimate) %>%
  arrange(`Follow-Up`) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 4
##   Index Disease_Group      `Follow-Up`      Estimate
##   <chr> <chr>          <chr>          <dbl>
## 1 6a    Psoriasis        "10 years"      0.74
## 2 6b    Psoriatic Arthritis (PsA) "10 years"      0.7
## 3 6c    Rheumatoid Arthritis (RA) "10 years"      0.716
## 4 7     Systemic Lupus Erythematosus (SLE) "10 years"      0.703
## 5 8a    Rheumatoid Arthritis (RA) "10 years"      0.752
## 6 8b    Ankylosing Spondylitis (AS) "10 years"      0.794
## 7 8c    Psoriatic Arthritis (PsA) "10 years"      0.764
## 8 8d    Psoriasis        "10 years"      0.815
## 9 19b   Diabetes (T2DM)      "10 years"      0.664
## 10 26b  Systemic Lupus Erythematosus (SLE) "10 years"      0.67
## 11 29b  Ankylosing Spondylitis (AS) "10 years"      0.66
## 12 35    Psoriatic Arthritis (PsA) "10 years"      0.866
## 13 1     Diabetes (T2DM)      "5 years"        0.663
## 14 36    Diabetes (T2DM)      "5 years (4.9 years)" 0.674
## 15 39    Rheumatoid Arthritis (RA) "6.0 years"      0.744
## 16 3     Systemic Lupus Erythematosus (SLE) "7 years (IQR 3-9 years)" 0.76
## 17 16    Diabetes (T2DM)      "8.68 ± 1.86years." 0.878
## 18 40    Rheumatoid Arthritis (RA) "9,957 patient-years of fo~ 0.79
## 19 23    Diabetes (T2DM)      "Up to 5 years"    0.72
```

```
library(dplyr)
library(meta)

fu_under10 <- c("1", "36", "3", "16", "40", "23", "39")

df_primary_logit <- df_primary_logit %>%
  mutate(Follow_group = ifelse(Index %in% fu_under10, "Under 10y", "10y or more"))

table(df_primary_logit$Follow_group, useNA = "ifany")
```

```
##
## 10y or more    Under 10y
##           12           7
```

```
sens_followup <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = df_primary_logit,
  sm = "PLOGIT",
```

```

method.tau = "REML",
method.random.ci = "HK",
subgroup = Follow_group,
tau.common = TRUE,
prediction = TRUE
)
summary(sens_followup)

```

```

##      proportion      95% CI %W(common) %W(random) Follow_group
## 1      0.6630 [0.6453; 0.6803]      2.8      6.3      Under 10y
## 3      0.7600 [0.5001; 0.9093]      0.0      1.4      Under 10y
## 6a     0.7400 [0.7195; 0.7595]      1.6      6.2     10y or more
## 6b     0.7000 [0.6361; 0.7569]      0.2      5.3     10y or more
## 6c     0.7160 [0.6904; 0.7403]      1.1      6.2     10y or more
## 7      0.7030 [0.5897; 0.7958]      0.1      3.9     10y or more
## 8a     0.7520 [0.7296; 0.7731]      1.3      6.2     10y or more
## 8b     0.7940 [0.7694; 0.8166]      0.8      6.1     10y or more
## 8c     0.7640 [0.7379; 0.7883]      0.9      6.1     10y or more
## 8d     0.8150 [0.7911; 0.8367]      0.8      6.0     10y or more
## 16     0.8780 [0.8317; 0.9129]      0.1      4.7      Under 10y
## 19b    0.6640 [0.6600; 0.6680]     53.7      6.4     10y or more
## 23     0.7200 [0.6995; 0.7396]      1.7      6.2      Under 10y
## 26b    0.6700 [0.6287; 0.7088]      0.5      5.9     10y or more
## 29b    0.6600 [0.4487; 0.8224]      0.0      2.2     10y or more
## 35     0.8660 [0.7660; 0.9273]      0.0      2.9     10y or more
## 36     0.6740 [0.6690; 0.6790]     33.3      6.4      Under 10y
## 39     0.7440 [0.7128; 0.7729]      0.7      6.0      Under 10y
## 40     0.7900 [0.7468; 0.8276]      0.3      5.6      Under 10y
##
## Number of studies: k = 19
##
##      proportion      95% CI      z|t      p-value
## Common effect model      0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK) 0.7430 [0.7087; 0.7745] 12.94 < 0.0001
## Prediction interval      [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1091 [0.0536; 0.2986]; tau = 0.3304 [0.2315; 0.5465]
## I^2 = 96.1% [94.4%; 98.9%]; H = 5.03 [4.21; 9.71]
##
## Test of heterogeneity:
##      Q d.f.      p-value
## 432.04 18 < 0.0001
##
## Results for subgroups (common effect model):
##      k proportion      95% CI      Q      I^2
## Follow_group = Under 10y      7      0.6785 [0.6739; 0.6831] 100.22 94.0%
## Follow_group = 10y or more 12      0.6752 [0.6715; 0.6789] 330.61 96.7%
##

```

```
## Test for subgroup differences (common effect model):
##           Q d.f.  p-value
## Between groups   1.21    1   0.2720
## Within groups  430.83   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##           k proportion      95% CI  tau^2    tau
## Follow_group = Under 10y    7    0.7467 [0.6625; 0.8157] 0.1091 0.3304
## Follow_group = 10y or more 12    0.7412 [0.6997; 0.7788] 0.1091 0.3304
##
## Test for subgroup differences (random effects model (HK)):
##           Q d.f.  p-value
## Between groups   0.02    1   0.8815
## Within groups  430.83   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

DESCRIPTIVE ANALYSIS - HELLERS K

```
library(dplyr)

carma <- df_all %>%
  filter(Index %in% c("5a", "5b", "5c")) %>%
  select(Index, Disease_Group, Country, Sample_size,
         Estimate, LowerCI, UpperCI, Dscrimination, `Follow-Up`)

carma

## # A tibble: 3 x 9
##   Index Disease_Group Country Sample_size Estimate LowerCI UpperCI Dscrimination
##   <chr> <chr>          <chr>         <dbl>    <dbl>    <dbl>    <dbl> <chr>
## 1 5a    "\r\n Rheuma~ Spain         708    0.607      NA      NA Heller's K
## 2 5b    "Ankylosing ~ Spain         692    0.592      NA      NA Heller's K
## 3 5c    "Psoriatic A~ Spain         680    0.58      NA      NA Heller's K
## # i 1 more variable: `Follow-Up` <chr>
```

SENSITIVITY ANALYSIS - CROSS-SECTIONAL

```
library(dplyr)

# Pulling the candidate cross-sectional surrogate-outcome studies from df_all
```

```
cross_sec <- df_all %>%
  filter(Outcome_type != "Appropriate") %>%
  select(Index, Disease_Group, Outcome_type, Outcome_Description,
         `Follow-Up`, Estimate, LowerCI, UpperCI, Dscrimination)

cross_sec
```

```
## # A tibble: 9 x 9
##   Index Disease_Group Outcome_type Outcome_Description `Follow-Up` Estimate
##   <chr> <chr>          <chr>          <chr>          <chr>          <dbl>
## 1 4      Diabetes (T1DM & ~ Somewhat_Ap~ "CVD( not includin~ "1 - 12 ye~    0.788
## 2 14     Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "3.3 years"  0.754
## 3 18     Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~  0.765
## 4 21     Metabolic Syndrom~ Not_Appropr~ "elevated cfPWV"    "Cross-sec~  0.697
## 5 22     Inflammatory Bowe~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~  0.812
## 6 27b    Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~  0.82
## 7 28     Rheumatoid Arthri~ Not_Appropr~ "carotid plaque fo~ "cross-sec~  0.64
## 8 32     chronic kidney di~ Not_Appropr~ "HbA1c and carotid~ "cross-sec~  0.51
## 9 37     Psoriatic Arthrit~ Not_Appropr~ "carotid plaque fo~ "cross-sec~  0.69
## # i 3 more variables: LowerCI <dbl>, UpperCI <dbl>, Dscrimination <chr>
```

```
library(dplyr)

cross_sec <- df_all %>%
  filter(Index %in% c("14", "18", "21", "22", "28", "32", "37", "27b")) %>%
  select(Index, Disease_Group, Outcome_Description, Dscrimination,
         Estimate, LowerCI, UpperCI, `Follow-Up`, Confidence_interval)

cross_sec
```

```
## # A tibble: 8 x 9
##   Index Disease_Group Outcome_Description Dscrimination Estimate LowerCI UpperCI
##   <chr> <chr>          <chr>          <chr>          <dbl> <dbl> <dbl>
## 1 14     Systemic Lup~ carotid plaque for~ AUC          0.754 0.64 0.87
## 2 18     Systemic Lup~ carotid plaque for~ AUC          0.765 0.711 0.82
## 3 21     Metabolic Sy~ elevated cfPWV      AUC          0.697 0.659 0.736
## 4 22     Inflammatory~ carotid plaque for~ AUC          0.812 0.748 0.875
## 5 27b    Systemic Lup~ carotid plaque for~ AUC          0.82 0.76 0.89
## 6 28     Rheumatoid A~ carotid plaque for~ AUC          0.64 0.53 0.75
## 7 32     chronic kidn~ HbA1c and carotid ~ AUC          0.51 0.32 0.7
## 8 37     Psoriatic Ar~ carotid plaque for~ AUC          0.69 0.6 0.77
## # i 2 more variables: `Follow-Up` <chr>, Confidence_interval <chr>
```

```
# Discrimination for subclinical carotid atherosclerosis
library(dplyr)
library(meta)

plaque <- df_all %>%
  filter(Index %in% c("14", "18", "22", "28", "37", "27b")) %>% # carotid plaque only; 21 &
  ↪ 32 excluded
  mutate(
    logit_c = log(Estimate / (1 - Estimate)),
```

```

    logit_lb = log(LowerCI / (1 - LowerCI)),
    logit_ub = log(UpperCI / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96)
)

nrow(plaque)

```

```
## [1] 6
```

```

pool_plaque <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = plaque,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  prediction = TRUE
)
summary(pool_plaque)

```

```

##      proportion      95% CI %W(common) %W(random)
## 14      0.7540 [0.6124; 0.8560]         6.9      11.4
## 18      0.7650 [0.7052; 0.8158]        31.8      21.2
## 22      0.8120 [0.7377; 0.8690]        16.4      17.4
## 27b     0.8200 [0.7403; 0.8793]        13.7      16.2
## 28      0.6400 [0.5215; 0.7436]        12.6      15.6
## 37      0.6900 [0.5984; 0.7688]        18.7      18.2
##
## Number of studies: k = 6
##
##              proportion      95% CI    z|t  p-value
## Common effect model      0.7542 [0.7206; 0.7850] 12.66 < 0.0001
## Random effects model (HK)  0.7527 [0.6747; 0.8171]  7.45   0.0007
## Prediction interval      [0.5719; 0.8740]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0804 [0.0000; 0.7589]; tau = 0.2835 [0.0000; 0.8711]
## I^2 = 60.2% [2.2%; 83.8%]; H = 1.58 [1.01; 2.48]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 12.55   5  0.0280
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 5)
## - Prediction interval based on t-distribution (df = 5)
## - Logit transformation

```

```
df_primary_logit %>%
  select(Index, Sample_size, Study_Year, `Follow-Up`
) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 4
##   Index Sample_size Study_Year `Follow-Up`
##   <chr>      <dbl>      <dbl> <chr>
## 1 1          20793        2025 "5 years"
## 2 3           143        2025 "7 years (IQR 3-9 years)"
## 3 6a          8062        2025 "10 years"
## 4 6b           769        2025 "10 years"
## 5 6c          4772        2025 "10 years"
## 6 7           550        2025 "10 years"
## 7 8a          91750       2023 "10 years"
## 8 8b          42306       2023 "10 years"
## 9 8c          26375       2023 "10 years"
## 10 8d         390664       2023 "10 years"
## 11 16           566        2022 "8.68 ± 1.86years."
## 12 19b        168871       2022 "10 years"
## 13 23          6245        2021 "Up to 5 years"
## 14 26b         1887        2021 "10 years"
## 15 29b         133        2020 "10 years"
## 16 35          155        2018 "10 years"
## 17 36        181399       2018 "5 years (4.9 years)"
## 18 39         12747       2017 "6.0 years"
## 19 40          1050       2013 "9,957 patient-years of follow-up\r\n9.5 years"
```

SUBGROUP ANALYSIS - MULTI-VS-SINGLE

```
library(meta)

# First check how the groups split and that the labels are clean
table(df_primary_logit$`Single_vs_multi-centre`, useNA = "ifany")
```

```
##
## Multi-centre Single-centre
##           14           5
```

```
primary_subgroup_centre <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = df_primary_logit,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  subgroup = `Single_vs_multi-centre`,
  tau.common = TRUE,
  prediction = TRUE
```

```
)
summary(primary_subgroup_centre)
```

```
##      proportion      95% CI %W(common) %W(random) Single_vs_multi-centre
## 1      0.6630 [0.6453; 0.6803]      2.8      6.3      Multi-centre
## 3      0.7600 [0.5001; 0.9093]      0.0      1.4      Single-centre
## 6a     0.7400 [0.7195; 0.7595]      1.6      6.2      Multi-centre
## 6b     0.7000 [0.6361; 0.7569]      0.2      5.3      Multi-centre
## 6c     0.7160 [0.6904; 0.7403]      1.1      6.2      Multi-centre
## 7      0.7030 [0.5897; 0.7958]      0.1      3.9      Single-centre
## 8a     0.7520 [0.7296; 0.7731]      1.3      6.2      Multi-centre
## 8b     0.7940 [0.7694; 0.8166]      0.8      6.1      Multi-centre
## 8c     0.7640 [0.7379; 0.7883]      0.9      6.1      Multi-centre
## 8d     0.8150 [0.7911; 0.8367]      0.8      6.0      Multi-centre
## 16     0.8780 [0.8317; 0.9129]      0.1      4.7      Single-centre
## 19b    0.6640 [0.6600; 0.6680]     53.7      6.4      Multi-centre
## 23     0.7200 [0.6995; 0.7396]      1.7      6.2      Single-centre
## 26b    0.6700 [0.6287; 0.7088]      0.5      5.9      Single-centre
## 29b    0.6600 [0.4487; 0.8224]      0.0      2.2      Multi-centre
## 35     0.8660 [0.7660; 0.9273]      0.0      2.9      Multi-centre
## 36     0.6740 [0.6690; 0.6790]     33.3      6.4      Multi-centre
## 39     0.7440 [0.7128; 0.7729]      0.7      6.0      Multi-centre
## 40     0.7900 [0.7468; 0.8276]      0.3      5.6      Multi-centre
##
## Number of studies: k = 19
##
##      proportion      95% CI      z|t      p-value
## Common effect model      0.6765 [0.6737; 0.6794] 110.10      0
## Random effects model (HK) 0.7430 [0.7087; 0.7745] 12.94 < 0.0001
## Prediction interval      [0.5926; 0.8518]
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1005 [0.0504; 0.2725]; tau = 0.3170 [0.2246; 0.5220]
## I^2 = 95.8% [94.6%; 96.8%]; H = 4.90 [4.29; 5.59]
##
## Quantifying residual heterogeneity (with 95% CIs):
## tau^2 = 0.1080 [0.0534; 0.2986]; tau = 0.3286 [0.2312; 0.5464]
## I^2 = 95.8% [97.6%; 99.6%]; H = 4.90 [6.42; 15.03]
##
## Test of heterogeneity:
##      Q d.f.      p-value
## 432.04 18 < 0.0001
##
## Results for subgroups (common effect model):
##      k proportion      95% CI      Q
## Single_vs_multi-centre = Multi-centre 14 0.6754 [0.6725; 0.6783] 372.44
## Single_vs_multi-centre = Single-centre 5 0.7199 [0.7028; 0.7364] 35.56
##      I^2
## Single_vs_multi-centre = Multi-centre 96.5%
## Single_vs_multi-centre = Single-centre 88.8%
##
## Test for subgroup differences (common effect model):
##      Q d.f.      p-value
```

```
## Between groups 24.04 1 < 0.0001
## Within groups 408.00 17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##
##           k proportion      95% CI  tau^2
## Single_vs_multi-centre = Multi-centre 14 0.7411 [0.7045; 0.7747] 0.1080
## Single_vs_multi-centre = Single-centre  5 0.7503 [0.6105; 0.8520] 0.1080
##
##           tau
## Single_vs_multi-centre = Multi-centre 0.3286
## Single_vs_multi-centre = Single-centre 0.3286
##
## Test for subgroup differences (random effects model (HK)):
##
##           Q d.f. p-value
## Between groups 0.04 1 0.8466
## Within groups 408.00 17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

```
data.frame(
  Q = primary_subgroup_centre$Q.b.random,
  df = primary_subgroup_centre$df.Q.b,
  p = primary_subgroup_centre$pval.Q.b.random
)
```

```
##           Q df      p
## 1 0.03743968 1 0.8465726
```

```
data.frame(
  centre = primary_subgroup_centre$subgroup.levels,
  k       = primary_subgroup_centre$k.w,
  c_stat = plogis(primary_subgroup_centre$TE.random.w),
  lower   = plogis(primary_subgroup_centre$lower.random.w),
  upper   = plogis(primary_subgroup_centre$upper.random.w),
  I2      = primary_subgroup_centre$I2.w
)
```

```
##           centre k  c_stat  lower  upper  I2
## Multi-centre Multi-centre 14 0.7411425 0.7044840 0.7747079 0.9650948
## Single-centre Single-centre  5 0.7502852 0.6105262 0.8520465 0.8875074
```


METAREGRESSION

Event meta regression

```
library(metafor)

# Log-transform events
df_primary_logit$log_events <- log(df_primary_logit$Events)

# Refit the primary model in metafor as the reference
m_base <- rma(
  yi = logit_c,
  sei = se_logit,
  data = df_primary_logit,
  method = "REML")
m_base

##
## Random-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of total heterogeneity): 0.1005 (SE = 0.0394)
## tau (square root of estimated tau^2 value):      0.3170
## I^2 (total heterogeneity / total variability):   98.67%
## H^2 (total variability / sampling variability):  75.42
##
## Test for Heterogeneity:
## Q(df = 18) = 432.0370, p-val < .0001
##
## Model Results:
##
## estimate      se      zval      pval      ci.lb      ci.ub
##  1.0615  0.0802  13.2316  <.0001   0.9043   1.2187  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Univariable meta-regression on log(events)
mr_events <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ log_events,
  data = df_primary_logit,
  method = "REML")
mr_events

##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):      0.1053 (SE = 0.0423)
## tau (square root of estimated tau^2 value):             0.3245
```

```
## I^2 (residual heterogeneity / unaccounted variability): 98.55%
## H^2 (unaccounted variability / sampling variability): 69.03
## R^2 (amount of heterogeneity accounted for): 0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 292.0704, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3616, p-val = 0.5476
##
## Model Results:
##
##          estimate      se      zval      pval      ci.lb      ci.ub
## intrcpt          1.2147  0.2664   4.5601 <.0001    0.6926   1.7368 ***
## log_events      -0.0226  0.0375  -0.6014  0.5476   -0.0962   0.0510
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mr_events
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity): 0.1053 (SE = 0.0423)
## tau (square root of estimated tau^2 value): 0.3245
## I^2 (residual heterogeneity / unaccounted variability): 98.55%
## H^2 (unaccounted variability / sampling variability): 69.03
## R^2 (amount of heterogeneity accounted for): 0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 292.0704, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3616, p-val = 0.5476
##
## Model Results:
##
##          estimate      se      zval      pval      ci.lb      ci.ub
## intrcpt          1.2147  0.2664   4.5601 <.0001    0.6926   1.7368 ***
## log_events      -0.0226  0.0375  -0.6014  0.5476   -0.0962   0.0510
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To make the slope interpretation on the c-statistic scale

```
# Predicted c-statistic across the range of events
# use the fitted metaregression model to estimate the c-statistic for studies at the
  ↪ event number
newpts <- data.frame(log_events = log(c(20, 100, 500, 5000, 25000))) # create dataframe
  ↪ for hypothetical moderators/event number
```

```

prediction <- predict( # function to calculate predicted values from fitted model
  mr_events, # fitted regression model
  newmods = as.matrix(newpts$log_events), # convert moderator to a matrix to feed to
  ↪ function
  transf = plogis
)

# build result table
data.frame(events = c(20, 100, 500, 5000, 25000),
  c_stat = round(prediction$pred, 3),
  lower = round(prediction$ci.lb, 3),
  upper = round(prediction$ci.ub, 3))

```

```

##   events c_stat lower upper
## 1     20  0.759 0.696 0.813
## 2    100  0.752 0.708 0.792
## 3     500  0.745 0.713 0.776
## 4    5000  0.735 0.693 0.774
## 5   25000  0.728 0.666 0.783

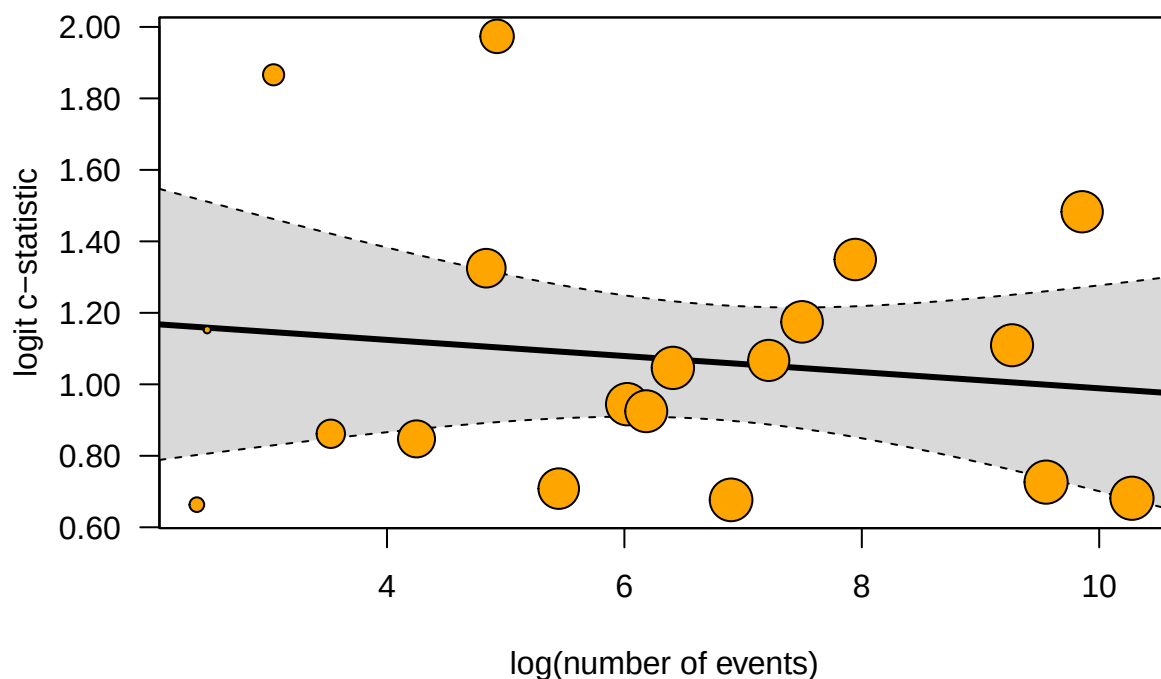
```

Bubble plot for the figure

```

regplot(mr_events,
  mod = "log_events", # name of relevant data in the model object
  xlab = "log(number of events)",
  ylab = "logit c-statistic",
  las = 1, # axis label
  digits = 2, # number of decimal places in the plot
  bg = "orange")

```



Event rate - meta regression

```
df_primary_logit$event_rate <- df_primary_logit$Events / df_primary_logit$Sample_size

mr_rate <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ event_rate,
  data = df_primary_logit,
  method = "REML")

mr_rate
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity): 0.1009 (SE = 0.0407)
## tau (square root of estimated tau^2 value): 0.3176
## I^2 (residual heterogeneity / unaccounted variability): 96.90%
## H^2 (unaccounted variability / sampling variability): 32.21
## R^2 (amount of heterogeneity accounted for): 0.00%
##
## Test for Residual Heterogeneity:
```

```
## QE(df = 17) = 372.6185, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 2.2794, p-val = 0.1311
##
## Model Results:
##
##           estimate      se    zval    pval    ci.lb    ci.ub
## intrcpt      0.8000  0.1910  4.1880 <.0001   0.4256  1.1743 ***
## event_rate    2.6616  1.7629  1.5098  0.1311  -0.7937  6.1168
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Sample Size - meta regression

```
df_primary_logit$log_n <- log(df_primary_logit$Sample_size)

mr_sample_size <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ log_n,
  data = df_primary_logit,
  method = "REML")

mr_sample_size
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):      0.1043 (SE = 0.0419)
## tau (square root of estimated tau^2 value):             0.3230
## I^2 (residual heterogeneity / unaccounted variability): 98.68%
## H^2 (unaccounted variability / sampling variability):    75.73
## R^2 (amount of heterogeneity accounted for):             0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 316.0538, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.6025, p-val = 0.4376
##
## Model Results:
##
##           estimate      se    zval    pval    ci.lb    ci.ub
## intrcpt      1.3182  0.3398  3.8794  0.0001   0.6522  1.9841 ***
## log_n       -0.0279  0.0360 -0.7762  0.4376  -0.0985  0.0426
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

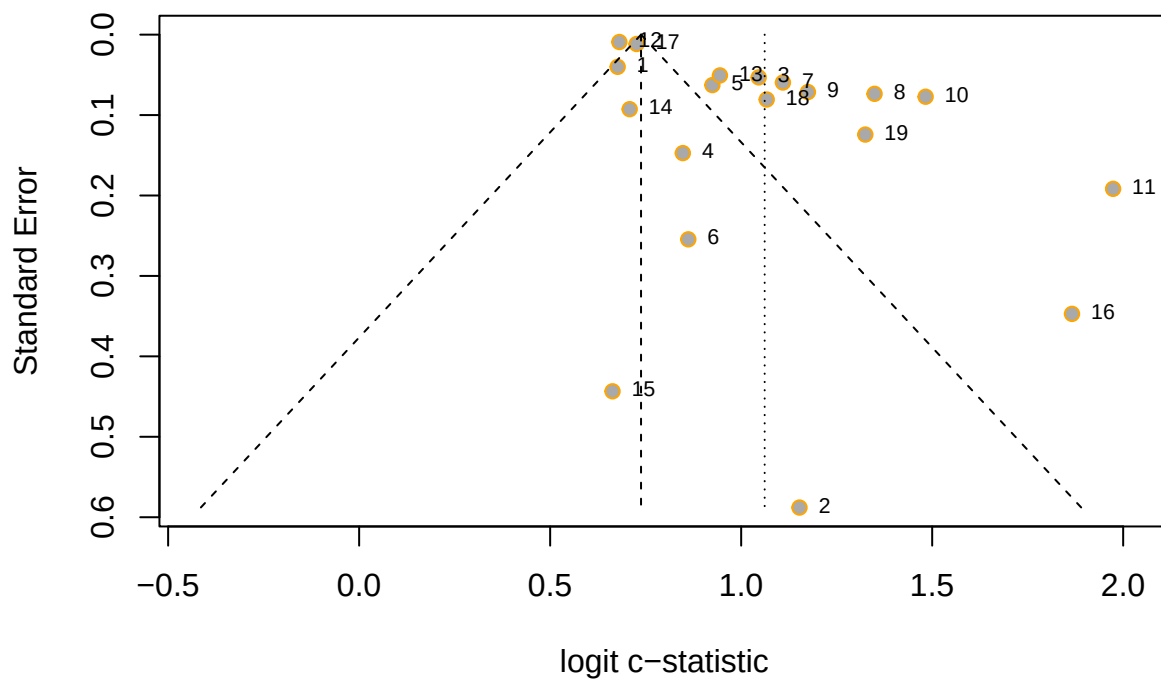
```

library(meta)

# Build the numbered key from the pooled object (row order = plot order)
key <- data.frame(
  No = seq_along(primary_logit$studlab), # create number sequence
  Study = primary_logit$studlab,
  c_stat = round(plogis(primary_logit$TE), 3) # study effect column
)

# Funnel plot labelled with numbers
meta::funnel(primary_logit,
  studlab = as.character(key$No), # display numbers
  cex.studlab = 0.7, # font size
  pos.studlab = 4, # label to the right of each point
  xlab = "logit c-statistic",
  backtransf = FALSE,
  col = "orange"
)

```



```

# Print the mapping to place in the figure caption or legend
print(key, row.names = FALSE)

```

```

## No Study c_stat
## 1 Enguita-German et al. (T2DM-2025) 0.663
## 2 Azoulay et al. (SLE-2025) 0.760

```

```
## 3      Hughes et al. (Ps-2025) 0.740
## 4      Hughes et al. (PsA-2025) 0.700
## 5      Hughes et al. (RA-2025) 0.716
## 6      Rosário et al. (SLE-2025) 0.703
## 7      Hughes et al. (RA-2023) 0.752
## 8      Hughes et al. (AS-2023) 0.794
## 9      Hughes et al. (PsA-2023) 0.764
## 10     Hughes et al. (Ps-2023) 0.815
## 11     Mu et al. (T2DM-2022) 0.878
## 12     Dziopa K et al. (T2DM-Q3-2022) 0.664
## 13     Hong et al. (T2DM-2021) 0.720
## 14     Sivakumaran et al. (SLE-Q3-2021) 0.670
## 15     Navarini et al. (AS-Q3-2020) 0.660
## 16     Navarini et al. (PsA-2018) 0.866
## 17     Read et al. (T2DM-2018) 0.674
## 18     Alemao et al. (RA-2017) 0.744
## 19     Arts EEA et al. (RA-2013) 0.790
```

```
# Egger-type test for funnel asymmetry
metabias(primary_logit, method.bias = "linreg", k.min = 10)
```

```
## Linear regression test of funnel plot asymmetry
##
## Test result: t = 4.62, df = 17, p-value = 0.0002
## Bias estimate: 4.4242 (SE = 0.9568)
##
## Details:
## - multiplicative residual heterogeneity variance (tau^2 = 11.2563)
## - predictor: standard error
## - weight: inverse variance
## - reference: Egger et al. (1997), BMJ
```

```
paste(paste0(key$No, " = ", key$Study), collapse = "; ") # caption for keys
```

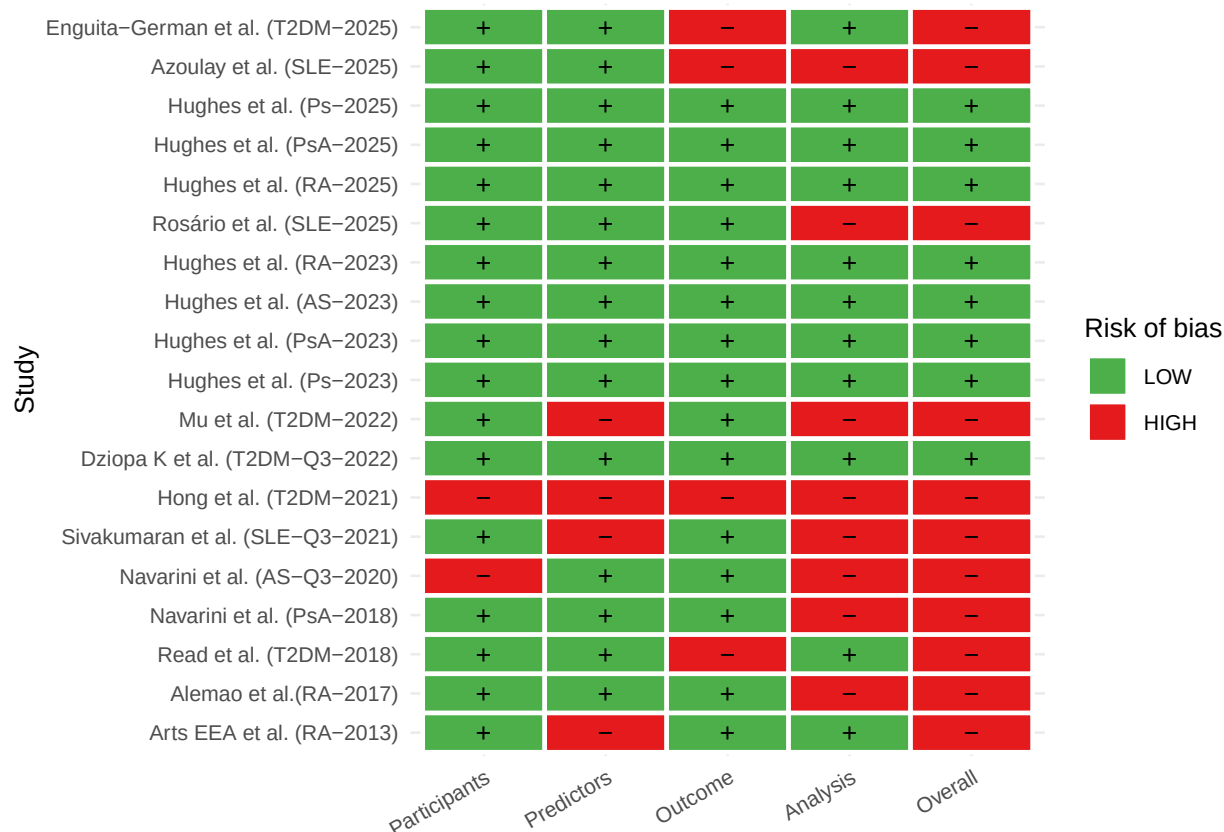
```
## [1] "1 = Enguita-German et al. (T2DM-2025); 2 = Azoulay et al. (SLE-2025); 3 = Hughes et al. (Ps-2025)"
```

```
#traffic light plot
library(dplyr); library(tidyr); library(ggplot2)

rob <- df_primary_logit %>%
  select(Index, Author, Participants, Predictors2, Outcome3, Analysis, Overall) %>%
  pivot_longer(c(-Index, -Author), names_to = "Domain", values_to = "Rating") %>%
  mutate(
    Domain = recode(Domain, Predictors2 = "Predictors", Outcome3 = "Outcome"),
    Domain = factor(Domain,
      levels = c("Participants", "Predictors", "Outcome", "Analysis", "Overall")),
    Rating = factor(toupper(trimws(Rating)), levels = c("LOW", "UNCLEAR", "HIGH"))

ggplot(rob, aes(x = Domain, y = factor(Author, levels = rev(unique(Author))))) +
  geom_tile(aes(fill = Rating), colour = "white", linewidth = 0.8) +
  geom_text(aes(label = recode(as.character(Rating),
    LOW = "+", UNCLEAR = "?", HIGH = "\u2212")), size = 3.5) +
```

```
scale_fill_manual(values = c(LOW = "#4DAF4A", UNCLEAR = "#FFD92F", HIGH = "#E41A1C")) +
labs(x = NULL, y = "Study", fill = "Risk of bias") +
theme_minimal(base_size = 10) +
theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



```
png("metareg_panel.png", width = 2400, height = 900, res = 200)
par(mfrow = c(1, 3), mar = c(5, 5, 3, 2))
regplot(mr_events, mod = "log_events", xlab = "log(events)",
        ylab = "logit c-statistic", main = "A", bg = "grey")
regplot(mr_sample_size, mod = "log_n", xlab = "log(sample size)",
        ylab = "", main = "B", bg = "grey")
regplot(mr_rate, mod = "event_rate", xlab = "Event rate",
        ylab = "", main = "C", bg = "grey")
dev.off()
```

```
## pdf
## 2
```

```
pkgs <- c("meta", "metafor", "metamisc", "readxl", "writexl",
          "dplyr", "stringr", "tidyr", "ggplot2", "gt")

purpose <- c(
  meta = "primary pooling (metagen), subgroups, leave-one-out, funnel plot",
  metafor = "case-deletion diagnostics, Baujat plot, meta-regression (rma)",
```



```

metamisc = "reference implementation of the Debray framework (valmeta)",
readxl   = "data import",
writexl  = "data export",
dplyr    = "data preparation",
stringr  = "text parsing of the AUC field",
tidyr    = "data reshaping",
ggplot2  = "risk-of-bias figures",
gt       = "formatted tables"
)

env <- data.frame(
  Package = pkgs,
  Version = vapply(pkgs, function(p) as.character(packageVersion(p)), character(1)),
  Purpose = purpose[pkgs],
  row.names = NULL
)

cat("R version:", paste0(R.version$major, ".", R.version$minor), "\n")

```

```
## R version: 4.5.2
```

```
cat("Platform :", R.version$platform, "\n\n")
```

```
## Platform : aarch64-apple-darwin20
```

```
print(env, right = FALSE)
```

```

##   Package Version
## 1  meta      8.5.0
## 2 metafor    4.8.0
## 3 metamisc   0.4.0
## 4 readxl     1.5.0
## 5 writexl    1.5.4
## 6 dplyr      1.2.1
## 7 stringr    1.6.0
## 8 tidyr      1.3.2
## 9 ggplot2    4.0.3
## 10 gt        1.3.0
##   Purpose
## 1 primary pooling (metagen), subgroups, leave-one-out, funnel plot
## 2 case-deletion diagnostics, Baujat plot, meta-regression (rma)
## 3 reference implementation of the Debray framework (valmeta)
## 4 data import
## 5 data export
## 6 data preparation
## 7 text parsing of the AUC field
## 8 data reshaping
## 9 risk-of-bias figures
## 10 formatted tables

```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] readr_2.2.0      tidyr_1.3.2      ggplot2_4.0.3
## [4] metafor_4.8-0    numDeriv_2016.8-1.1 metadat_1.4-0
## [7] Matrix_1.7-4     metamisc_0.4.0    meta_8.5-0
## [10] metabook_0.2-0   writexl_1.5.4     stringr_1.6.0
## [13] dplyr_1.2.1      readxl_1.5.0
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.6      generics_0.1.4    xml2_1.5.2        stringi_1.8.7
## [5] lattice_0.22-7  pROC_1.19.0.1     hms_1.1.4         lme4_2.0-1
## [9] digest_0.6.39   magrittr_2.0.4    evaluate_1.0.5     grid_4.5.2
## [13] RColorBrewer_1.1-3 mvtnorm_1.3-3     CompQuadForm_1.4.4 fastmap_1.2.0
## [17] plyr_1.8.9      cellranger_1.1.0  purrr_1.2.1        scales_1.4.0
## [21] reformulas_0.4.4 Rdpack_2.6.6      cli_3.6.5          rlang_1.1.7
## [25] rbibutils_2.4.1  splines_4.5.2     withr_3.0.2        yaml_2.3.12
## [29] otel_0.2.0      tools_4.5.2       tzdb_0.5.0         nloptr_2.2.1
## [33] minqa_1.2.8     mathjaxr_2.0-0    boot_1.3-32        vctr_0.7.1
## [37] R6_2.6.1        lifecycle_1.0.5   MASS_7.3-65        pkgconfig_2.0.3
## [41] pillar_1.11.1   gtable_0.3.6      glue_1.8.0         Rcpp_1.1.1
## [45] xfun_0.56       tibble_3.3.1      tidyrselect_1.2.1  rstudioapi_0.18.0
## [49] knitr_1.51      farver_2.1.2      htmltools_0.5.9    nlme_3.1-168
## [53] labeling_0.4.3  rmarkdown_2.30    compiler_4.5.2     S7_0.2.1
```

```
citation()      # R itself
```

```
## To cite R in publications use:
##
## R Core Team (2025). _R_: A Language and Environment for Statistical
## Computing. R Foundation for Statistical Computing, Vienna, Austria.
## <https://www.R-project.org/>.
##
## A BibTeX entry for LaTeX users is
##
```

```
## @Manual{,
##   title = {R: A Language and Environment for Statistical Computing},
##   author = {{R Core Team}},
##   organization = {R Foundation for Statistical Computing},
##   address = {Vienna, Austria},
##   year = {2025},
##   url = {https://www.R-project.org/},
## }
##
## We have invested a lot of time and effort in creating R, please cite it
## when using it for data analysis. See also 'citation("pkgname")' for
## citing R packages.
```

```
citation("meta")
```

```
## To cite package 'meta' in publications use:
##
## Balduzzi S, Rücker G, Schwarzer G (2019), How to perform a
## meta-analysis with R: a practical tutorial, Evidence-Based Mental
## Health; 22: 153-160.
##
## A BibTeX entry for LaTeX users is
##
## @Article{,
##   title = {How to perform a meta-analysis with {R}: a practical tutorial},
##   author = {Sara Balduzzi and Gerta Rücker and Guido Schwarzer},
##   journal = {Evidence-Based Mental Health},
##   year = {2019},
##   volume = {22},
##   pages = {153--160},
## }
##
## DOI: 10.1136/ebmental-2019-300117
```

```
citation("metafor")
```

```
## To cite the metafor package in publications, please use:
##
## Viechtbauer, W. (2010). Conducting meta-analyses in R with the
## metafor package. Journal of Statistical Software, 36(3), 1-48.
## https://doi.org/10.18637/jss.v036.i03
##
## A BibTeX entry for LaTeX users is
##
## @Article{,
##   title = {Conducting meta-analyses in {R} with the {metafor} package},
##   author = {Wolfgang Viechtbauer},
##   journal = {Journal of Statistical Software},
##   year = {2010},
##   volume = {36},
##   number = {3},
##   pages = {1--48},
```

```
##     doi = {10.18637/jss.v036.i03},  
##   }
```

```
citation("metamisc")
```

```
## To cite package 'metamisc' in publications use:
```

```
##
```

```
## Debray T, de Jong V (2022). _metamisc: Meta-Analysis of Diagnosis and  
## Prognosis Research Studies_. doi:10.32614/CRAN.package.metamisc  
## <https://doi.org/10.32614/CRAN.package.metamisc>, R package version  
## 0.4.0, <https://CRAN.R-project.org/package=metamisc>.
```

```
##
```

```
## A BibTeX entry for LaTeX users is
```

```
##
```

```
## @Manual{,
```

```
##   title = {metamisc: Meta-Analysis of Diagnosis and Prognosis Research Studies},
```

```
##   author = {Thomas Debray and Valentijn {de Jong}},
```

```
##   year = {2022},
```

```
##   note = {R package version 0.4.0},
```

```
##   url = {https://CRAN.R-project.org/package=metamisc},
```

```
##   doi = {10.32614/CRAN.package.metamisc},
```

```
## }
```